

Leveraging an October Surprise to Estimate Coattail Effects

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Abstract

Does a presidential candidate's popularity impact her party's down-ballot success? So-called "coattail effects" can be challenging to identify: it is difficult to distinguish between voters' opinion of a prominent candidate and her political party. To overcome this challenge, I exploit a shock to candidate popularity created by late-election information in the 2016 presidential election. Differences in states' convenience voting laws created variation in voters' ability to respond to FBI Director James Comey's letter to Congress. Results indicate that knowledge of the letter significantly increased Republican presidential vote share. Using an instrumented difference-in-differences model, I find that a one percentage point increase in Republican presidential vote share increases down-ballot Republican vote share by 0.79 percentage points.

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1 Introduction

Political parties act as a coordination mechanism through which voters are mobilized, resources raised, and policies passed in the name of advancing common ideological goals. They also lower informational costs to voters: rather than taking the time to evaluate every candidate each election, a voter can interpret partisan affiliation as a signal of a candidate’s values and objectives. This generates the possibility for electoral spillovers between candidates, especially from ones with high visibility to those with less. These spillovers are known as “coattail effects”: a candidate has coattail influence if down-ballot candidates in her same party receive votes they otherwise would not have received, save for her popularity [Miller, 1955]. Coattail effects, though widely believed to exist, can be challenging to quantify: the popularity of a single candidate is difficult to parse from the public’s sentiment toward her political party. In this paper, I provide the first causally identified estimates of presidential coattail effects.

I estimate presidential coattail effects using a natural experiment created by late-election information in the 2016 U.S. presidential election. On October 28, 2016, then-FBI Director James Comey sent a letter to Congress regarding Democratic presidential candidate Hillary Clinton. Comey announced the discovery of additional emails which pertained to an investigation into Clinton’s use of a private email server while serving as Secretary of State – an investigation he previously testified was complete. The letter reflected negatively on Clinton, but variation in early voting laws across states meant voters had differing abilities to react to the information. Many people in states with early voting had already cast their ballots. In my central identification strategy, I compare election outcomes in counties with no-excuse early voting available before 10/28 to those in counties where no-excuse early voting was either unavailable or began after 10/28, relative to election outcomes in 2012.

First, I show that the Comey letter impacted the 2016 presidential election: exposure to the letter decreased the Democratic presidential vote share by 2 percentage points and increased the Republican presidential vote share by 3.2 percentage points. This increase is larger than the vote margins of eight states in 2016: of these eight, four did not have early voting ahead of October 28th, and three of those four (Florida, Michigan, and Pennsylvania) broke in favor of Republican candidate Donald Trump.¹ Using this as a first stage, I use an instrumented difference-in-differences model to estimate coattail effects. I find that a one percentage point increase in the Republican presidential vote share led to a 0.79 percentage

¹Of the four that did have early voting ahead of the letter’s release, three (Minnesota, Nevada, and Maine) broke in favor of Clinton.

point increase in the down-ballot Republican vote share.

The magnitude of these results is politically significant. The 2017-2019 Senate was comprised of 51 Republicans and 49 Democrats, with Vice President Pence casting 13 tie-breaking votes, more than all tie-breaking votes cast in the previous 14 Senates combined [Schaeffer, 2024]. Pence cast tie-breaking votes to confirm the nominations of the secretary of education and the deputy director of the Office of Management and Budget, as well as on measures related to healthcare, consumer protections, and Title X funding for family planning and related services [Ballotpedia, c].² Using my estimated effect of a 3.2 p.p. decrease in Democratic down-ballot vote share, 2 Republican seats in states most able to respond to the Comey letter would have been Democratic in the absence of the letter, flipping control of the Senate. As such, the coattail effects identified in this paper not only had political effects, but direct policy effects as well.³

We might be concerned that these findings reflect differential political trends of early voting and non-early voting states rather than the impact of the Comey letter. To address these concerns, I show that other items on states' ballots are unrelated to the timing and availability of early voting with a balance test. To add further support for the parallel trends assumption, I leverage several high-frequency datasets to compare treatment and control states in the weeks leading up to Election Day in 2012 versus 2016 with triple-difference event studies. I find no significant pretrends using numerous measures of campaign activity and voters' intentions.

To provide additional support for my results, I limit my analysis to six early voting states which separately report votes cast on Election Day versus during the early voting period.⁴ This allows me to compare partisan vote shares of early voters and Election Day voters to assess the Comey effect. These states had varying amounts of early voting available ahead of the letter's release. I expect to observe larger effects in states that had more (vs. less) days of early voting available prior to 10/28, as those states presumably had a larger proportion of voters casting their ballot with no knowledge of the letter. I use a difference-in-differences design with continuous treatment where the number of days of early voting available before 10/28 is the treatment dose. I find that in a county with 10.17 days of early voting ahead of the Comey letter (the average number of days of early voting prior to 10/28 across all

²Pence's tie-breaking vote to confirm Secretary of Education DeVos was the first instance of a Vice President using this power in a Cabinet nomination vote in U.S. history.

³For example, the Tax Cuts and Jobs Act was passed using the ballot reconciliation process, with 51 Republicans voting in favor and 49 Democrats voting against it. See https://www.senate.gov/legislative/LIS/roll_call_votes/vote1151/vote_115_1_00303.htm.

⁴The six states are Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina.

states with no-excuse early voting in 2016), exposure to the letter decreased Democratic presidential vote share by 3.05 percentage points and increased Republican presidential vote share by 2.03 percentage points. These findings stem from a different source of variation but are quite similar to those from my main specification.

To better understand the mechanisms behind these results, I provide a stylized model of coattail effects with a negative shock to candidate valence. Formal theoretical treatments of coattail effects include Zudenkova [2011] and Halberstam and Montagnes [2015]. Zudenkova [2011] shows that such effects arise if we assume incumbents prefer their co-partisans to win races in different levels of government.⁵ Halberstam and Montagnes [2015] develop a model in which uninformed voters are assumed to be straight-ticket voters who align their down-ballot vote choice with their presidential one. These are low-propensity voters who vote in presidential elections but not midterm elections, generating ideological differences in senators elected in presidential and midterm years.⁶ Also relevant here is McCrary [2025], which formally models how voters respond to late-election information in the form of valence shocks, showing that some voters respond to negative information about their ideologically-preferred candidate by voting for her opponent or abstaining entirely.

I focus on the case of presidential elections in Halberstam and Montagnes’ model, adapting it to allow uninformed voters to abstain. Taking insights from McCrary [2025], I demonstrate how the Comey letter could have generated coattail effects via two distinct mechanisms. First, the letter may have decreased the proportion of uninformed voters who preferred the Democratic presidential candidate (the “vote-switching” channel). Second, the letter may have increased the proportion of uninformed voters who preferred the Democratic presidential candidate but chose to abstain (the turnout channel).

I formally test the turnout mechanism using daily counts of early votes cast by party registration in North Carolina. This unique system of reporting early voting turnout provides for a natural comparison between pre-October 28th and post-October 28th early voters in 2012 versus 2016 with a two-way fixed effects event study. I find that the letter decreased registered Democrats’ share of early turnout by 2.3 percentage points in North Carolina. This

⁵Knowing this, retrospective voters (meaning they vote based on an incumbent’s performance) jointly evaluate co-partisans, providing for two-sided coattails (that is, spillovers both up and down the ballot).

⁶The authors document that senators first elected in presidential years are more ideologically extreme than those first elected in midterms and devise the model to explain this observation. It should be noted that this pattern of differing senator ideologies may have other explanations. For example, it could be that parties run more extreme Senate candidates to generate additional excitement for the party in presidential years. The documented correlation is between senator ideology and the *presence* of a presidential election, not co-partisan presidential success, and thus could be a result of strategic choices by both parties in presidential years, independent of their presidential candidates’ popularity.

provides evidence that the Comey letter generated coattail effects for down-ballot Democrats by decreasing turnout among those who otherwise would have voted for them.

This is the first paper to leverage an October surprise to estimate coattail effects. Although there is a large literature devoted to estimating coattail effects, proper identification of presidential coattails is difficult to achieve. The strategy used here offers a new method of isolating a shock to top-ballot candidate popularity and allows for formal testing of the mechanisms behind coattail effects. Additionally, I provide the first quasi-experimental analysis of the Comey letter’s impact on the 2016 presidential election and concurrent down-ballot races. These findings are politically significant and have direct policy implications given the narrow Republican control of the 2017-2019 Senate. My findings imply that coattail effects can lead to more instances of unified government, which may affect both legislative efficacy and the types of policies passed.

The presence and magnitude of coattail effects also have significant strategic implications for politicians. While coattail effects are likely not a determinant of down-ballot candidates’ party affiliation,⁷ they may influence how closely such candidates choose to align themselves with up-ballot ones. Presidential coattail effects may even influence *when* down-ballot candidates run: they may wish to obtain an electoral boost from a popular co-partisan or avoid being dragged down by an unpopular one. Finally, this paper’s finding that late-election information can impact presidential and down-ballot races suggests that the incentives for the strategic release of late-election information may be even larger than previously thought [Gratton et al., 2018].

2 Background

2.1 Coattail Effects

Though there is a large literature on coattail effects, credible identification is difficult: there are inherent complications in separating a particular candidate’s popularity from that of

⁷Though there is a recent uptick in party-switching among state lawmakers in the U.S. [Crampton, 2023].

her political party.⁸ How might we distinguish coattail effects from party politics? Coattail effects are defined as the impact of a candidate’s personal popularity on her co-partisans’ success elsewhere on the ballot. Thus, proper identification of such an effect requires some exogenous shock to a candidate’s popularity – importantly, separate from the favorability of her party at large – and variation in voters’ experience of that shock.

Meredith [2013] exploits the excess support granted by (geographically) nearby voters to estimate gubernatorial coattail effects, finding that a one percentage point increase in gubernatorial vote share leads to 0.1-0.2 percentage point increases in vote shares for partisan allies running for secretary of state and attorney general. Many recent papers use close-election regression discontinuity designs to estimate the effect of a co-partisan winning a prior election in a different level of government. These have yielded mixed results on the existence and direction of such “incumbency spillovers,” depending on setting, the levels of government, and the direction [Hainmueller and Kern, 2008, Broockman, 2009, Folke and Snyder, 2012, Ade and Freier, 2013, Feierherd, 2020, Ventura, 2021, Savu, 2024].⁹

It is possible that each of these designs could be capturing not just an increase in the personal popularity of a candidate, but also an amplified opportunity for party rhetoric. Geographically proximate candidates and incumbents alike have increased opportunities to positively advertise their own political party, complicating the distinction between coattail effects and party politics. Instead, the Comey letter (or more broadly, late-election information) provides a plausibly exogenous shock to candidate popularity alone.

Late-election information offers the popularity shock needed to estimate presidential coattails, and disparate state-level early voting laws supply the necessary variation. Key to my empirical design is the notion that convenience voters and Election Day voters face fundamentally different information environments [Meredith and Malhotra, 2011]. My identification strategy is similar to ones used in Montalvo [2011], which shows that a terrorist bombing in Spain negatively impacted the performance of the incumbent party in the 2004 general election which took place three days later, and Graham and Svolik [2020], which shows that a Montana U.S. House candidate’s assault of a journalist the night before an

⁸Most prior work consists of observational studies which attempt to control for determinants of down-ballot vote shares [Campbell and Sumners, 1990, Mondak, 1993, Flemming, 1995, Mattei and Glasgow, 2005] or structural models [Kramer, 1971, Ferejohn and Calvert, 1984] which have proven to be sensitive to model specification [Fair, 2009]. See Meredith [2013] for a detailed discussion of the difficulties of identifying coattail effects.

⁹Several of these papers estimate “reverse coattails,” which refer to spillovers from down-ballot candidates to their fellow party members higher up on the ticket; for example, from a Democratic mayoral candidate to a Democratic U.S. House candidate.

election hurt his performance with moderate voters on Election Day. This paper is the first to use late-election information to 1) analyze its impact in a U.S. presidential race and 2) identify coattail effects.

The paper is also related to work on the impact of information and media in elections [Couttenier et al., 2024, Ash and Galletta, 2023, Wang, 2021, Morton et al., 2015, Chiang and Knight, 2011, DellaVigna and Kaplan, 2007]. Some work has been done on the impact of the Comey letter, specifically: there is some evidence that the letter had impacts in electoral prediction markets [Halcoussis et al., 2020]. McKee et al. [2019] points out that in each of the seven swing states that Trump won, he lost the sum of the votes cast early in-person or by mail; this paper builds on this by using variation in early voting to provide the first quasi-experimental analysis of the Comey letter.

Coattail effects can have important effects on voter welfare through increasing the likelihood of unified government: a popular president-elect is more likely to take office with partisan control of both congressional chambers. This may make voters worse off if they prefer that executive and legislative branches are controlled by opposing parties to moderate policy outcomes [Alesina and Rosenthal, 1996] or provide more effective oversight of the executive [Fox and Van Weelden, 2010]. At the same time, unified government may decrease legislative gridlock: Tsebelis [1995] shows that the likelihood of policy reform is decreasing in the number of political veto players. Indeed, studies find that under divided governments, budget approvals are delayed [Kirkland and Phillips, 2018] and legislators' ability to respond to revenue shocks is reduced [Alt and Lowry, 1994]. Empirical evidence on policy outcomes under divided governments is mixed: some studies have found that state legislators under divided governments implement more moderate policies [Repetto and Sosa Andrés, 2023] or spend more on bipartisan priorities such as infrastructure [Buchheim and Fretz, 2020]. However, Bernecker [2016] finds that in the US, state legislatures under divided governments are 25% *more* likely to adopt a welfare reform, theorizing that when state legislatures are controlled by the governor's opposition party, they can confront the governor with policy issues through passing bills. The large documented effects of unified versus divided governments, in combination with the magnitude of estimates found in this paper, highlight the importance of coattail effects in determining policy.

2.2 Convenience Voting in the United States

Thirty-five states began convenience voting before October 28th, 2016, the day of the Comey letter’s publication. A total of thirty-seven states practiced no-excuse convenience voting at the time: see Figure 1 for statewide variation (note that Florida and Oklahoma had no-excuse convenience voting, but their voting periods did not open until *after* October 28th). Of those thirty-seven, three states used universal vote-by-mail, meaning ballots were mailed to all registered voters.¹⁰ The rest of the thirty-seven had early in-person voting options, where citizens could cast their ballot in-person up to fifty days ahead of Election Day.¹¹

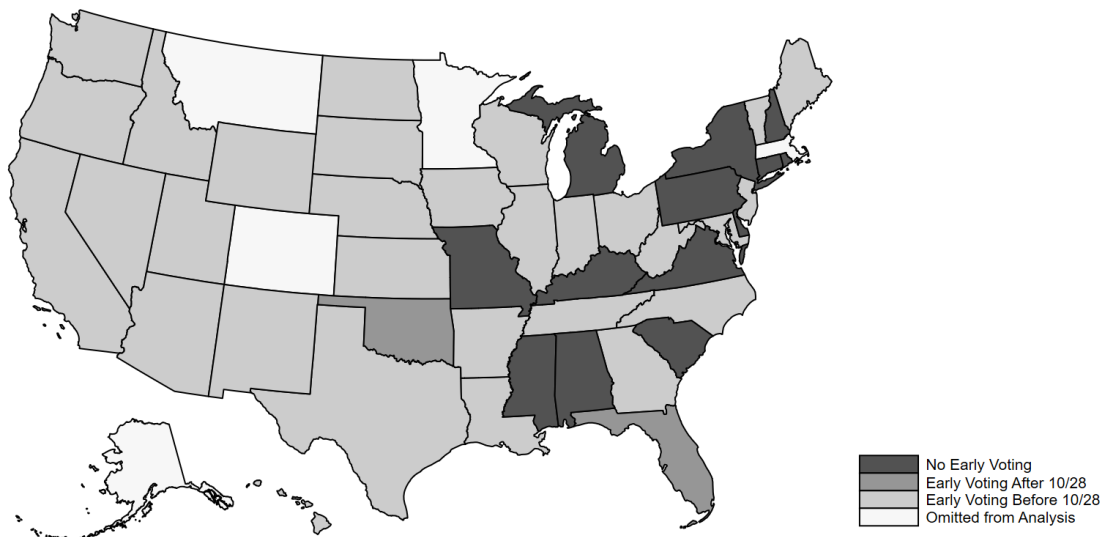


Figure 1: This map displays variation in convenience voting laws for U.S. states in 2016.

Notes: Four states (Colorado, Massachusetts, Minnesota, and Montana) made changes to their convenience voting laws between 2012 and 2016; I omit them from my analysis. I also omit Alaska, as it does not report election results by county.

See Table 1 for summary statistics on counties in states with convenience voting and states without convenience voting. A potential concern with comparing the two groups is that there may be underlying qualities which differ between the two and are somehow

¹⁰These three states were Colorado, Oregon, and Washington; nearly 80% of counties in Utah were also using universal vote-by-mail in 2016 [Thompson et al., 2020].

¹¹The earliest early voting start date in 2016 was September 19th in Wisconsin Rapids, Wisconsin.

correlated with treatment. That is, if one group was more or less responsive to the Comey letter due to differential baseline characteristics. However, as Table 1 shows, the two groups possess comparable baseline demographic, economic, and political attributes.

Table 1: This table displays summary statistics in 2012 for counties in the United States, broken into states with convenience voting and counties without convenience voting.

	Convenience voting	No convenience voting	Difference (std. devs.)
18+ population	78,171.43	77,744.84	0.002
Unemployment rate	7.55	8.74	0.424
Population density	176.72	519.7	0.188
Median age	40.26	40.28	0.004
Number of men per 100 women	100.83	98.33	0.221
Share non-white	15.43	18.46	0.183
Median income (in \$10,000s)	4.55	4.41	0.122
Share with Bachelor’s or higher	19.02	19.15	0.015
Share white without college degree	79.91	79.27	0.07
Dem/Rep. House margin*	-0.23	-0.17	0.186
Dem. vote share in pres. race	0.37	0.42	0.336
Turnout	0.55	0.57	0.193
Count	2090	801	

Notes: The means of the convenience voting and no convenience voting groups are reported, along with the difference in terms of the standard deviation of the entire sample, for reference. For example, the difference in 18+ population between the groups is 0.002 of one standard deviation (measured from the entire sample). The variable denoted with an asterisk is averaged over three U.S. election cycles: 2008, 2010, and 2014. All other variables are taken from 2012. Colorado, Massachusetts, Minnesota, and Montana are not included, as they changed convenience voting laws between 2012 and 2016; Alaska is not included, as it does not report election results by county. Population density is calculated by dividing a county’s 18+ population by its land area in square miles (as reported by the 2010 Census). The share with Bachelor’s or higher variable refers to the percent of those aged 25+ with a bachelor’s or more advanced degree.

2.3 The Comey Letter as Late-Election Information

Late-election information, broadly speaking, is revealed to voters in the weeks or days preceding Election Day and has the potential to change voters’ minds. This information can take the form of something already structured within the political system, such as presidential debates or campaign stops, or a spontaneous revelation, known as the “October surprise” in American politics. Interested parties certainly have incentives to release information with the intent to impact election outcomes as Election Day looms.¹² However, this late-election

¹²Gratton et al. [2018] present this as a tradeoff between credibility and scrutiny: good signals of candidate quality are released earlier, as this conveys credibility of the information, but poor signals will be released later, since they would not hold up to the public’s scrutiny if examined for too long.

information, whether spontaneous or strategically released, can nevertheless “surprise” the public and might induce changes in voter behavior.

There are many recent examples of late-election information in the United States: in 2020, news of an extramarital affair and a subsequent investigation by the U.S. Army Reserve of Cal Cunningham, a Democrat running for a U.S. Senate seat in North Carolina, occurred less than one month away from Election Day [Robertson, 2020]. In the 2003 California gubernatorial recall election, Republican Arnold Schwarzenegger replaced incumbent Democratic Governor Gray Davis, winning by a margin of 968,491 votes [Ballotpedia, d]. The *Los Angeles Times* reported on the Friday before the election that six women had accused Schwarzenegger of sexual misconduct [Cohn et al., 2003] after “more than three million of ten million voters had already cast their ballots” [Gronke et al., 2008]. In 2000, news of George W. Bush’s 1976 DUI arrest was reported on by media outlets four days before Election Day [Balz, 2000].

In this paper, I focus on a particular example of late-election information: then-FBI Director James Comey’s October 28, 2016 letter to Congress. Comey announced in the letter that the FBI had discovered more emails which seemed pertinent to the investigation of Hillary Clinton’s use of a private email server during her time as Secretary of State (an investigation Comey had previously testified was complete). The letter was sent to Congress just eleven days before the presidential election between Republican candidate Donald Trump and Democratic candidate Hillary Clinton [Perez and Brown, 2016].

The last month or so of the campaign leading up to Election Day on November 8th was quite dynamic in terms of late-election information. On October 7th, *The Washington Post* obtained a video in which “Donald Trump bragged in vulgar terms about kissing, groping and trying to have sex with women during a 2005 conversation caught on a hot microphone” [Fahrenthold, 2016]. This prompted several prominent Republicans to condemn Trump; some even suggested he withdraw his candidacy [Wellford, 2016]. On that same day, the Office of the Director of National Intelligence and the Department of Homeland Security released a statement accusing Russia of interfering with the election, and media organization WikiLeaks began releasing emails belonging to Clinton campaign chairman John Podesta, many of them “embarrassing for Clinton” [Cohen, 2017]. These releases continued on a near-daily basis throughout the rest of the campaign, but none seemed to generate a nation-wide shock in the way that 1) the aforementioned tape, 2) the beginning of the WikiLeaks releases, or 3) the Comey letter did.

I choose to focus on the Comey letter as the late-election information of interest in the election. The events of October 7th negatively impacted both candidates, prompting muddled behavioral implications for voters. In comparison, the Comey letter (and thus October 28th) clearly represented negative information for Clinton alone. Additionally, not as many people had yet had the opportunity to vote by October 7th, making for a smaller control group: just ten states began convenience voting before October 7th.¹³

As mentioned above, thirty-five states began convenience voting before October 28th. Although concerns over Clinton’s use of the email server had already been raised and an investigation completed, this letter and its implication of a re-opened investigation may have legitimized some voter’s concerns about the candidate. It certainly represented an information shock of interest to many voters; see Figure 2 in the appendix for Google Trends data, which shows a large spike in searches for “Comey” on 10/28/16. Election Day was still eleven days away, yet many had already cast their ballot before they learned of this investigation. I leverage this variation in voters’ knowledge of the letter at the time of casting their ballot, plausibly exogenous due to differences in states’ early voting policies, to provide causally identified estimates of the letter’s effect on election outcomes in 2016.

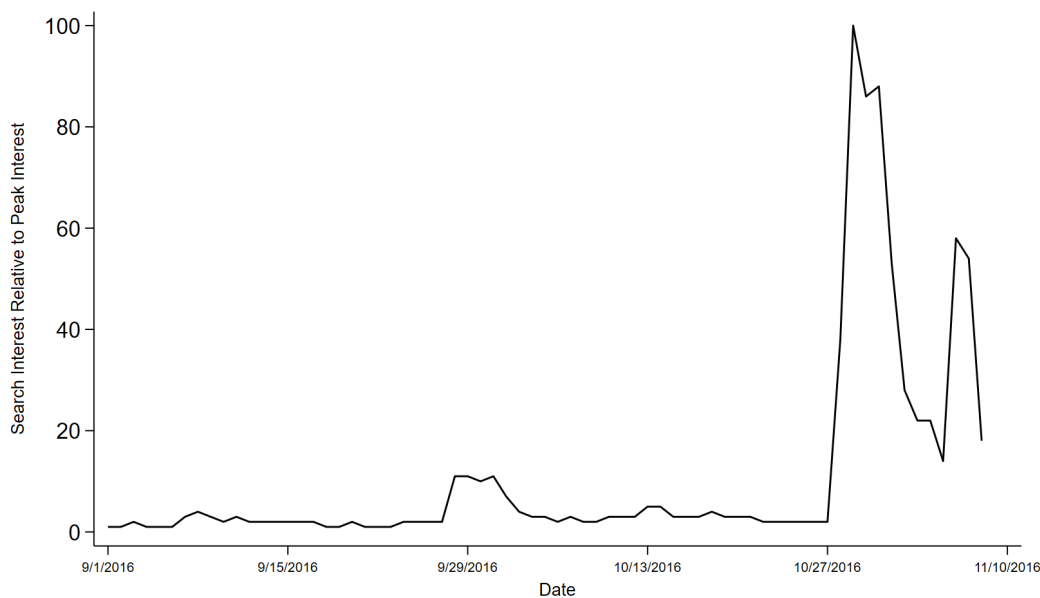


Figure 2: This figure shows Google Trends data for searches for “Comey” in the United States between 9/1/2016-11/8/2016. A value of 100 is peak popularity for the term.

¹³In a robustness check, I show that results in this paper do not hold if treatment is redefined around early voting availability ahead of October 7th, emphasizing the conflicting effects the numerous events of the day may have had on voters: see Section A.2.4.

3 Theoretical Framework

The Comey letter generated a negative shock to Clinton’s popularity. McCrary [2025] formally models how voters respond to late-election information in the form of a valence shock. In that model, voters value both candidates’ ideology and valence. Voting is costly, so negative valence shocks may lead to abstention from some voters. Three types of voter behavior arise endogenously from how voters respond to late-election information. Strong partisans (those with extreme ideological preferences) are unaffected by valence shocks and always vote for their ideologically-preferred candidate. Weak partisans (those with less extreme ideological preferences) would abstain in the face of sufficiently negative valence realizations for their ideologically-preferred candidate but would never choose to vote for the candidate they are ideologically further from. Finally, swing voters (those with the most moderate ideological preferences) would vote for the candidate they are ideologically further from in the presence of sufficiently negative valence shocks for their ideologically-preferred candidate (or positive ones for the other candidate).

In the context of the Comey letter, this model implies that some left-leaning voters would have voted for Trump or abstained entirely in response to the letter. This could have led to negative electoral spillovers for down-ballot Democrats. In the following section, I adopt the model of coattails developed by Halberstam and Montagnes [2015], incorporating insights from McCrary [2025], to highlight two mechanisms through which the Comey letter may have affected down-ballot candidate performance: preference shifts (“vote-switching”) and changes in turnout. First, I summarize the setup of their model.¹⁴

Assume there are two contemporaneous races (here, presidential and down-ballot – this could be a Senate race, a U.S. House race, gubernatorial election, etc.). Two candidates (a Democrat and a Republican) compete in each race. Suppose that candidates’ ideologies are independent of one another: in particular, Democratic presidential and down-ballot candidates’ ideologies are independent draws from a uniform distribution from -1 to 0 , and Republican presidential and down-ballot candidates’ ideologies are independent draws from a uniform distribution from 0 to 1 . There are many citizens, each of whom have some ideological bliss point, distributed uniformly between -1 and 1 .

There are two types of citizens: informed voters, who observe the ideological positions of all four candidates, and uninformed voters, who observe only the ideological positions

¹⁴Microfoundations of Halberstam and Montagnes’ model can be found in the Online Appendix of their paper.

of the presidential candidates. Let uninformed voters constitute a proportion $\delta > 0$ of the electorate. If voters know the ideological positions of the candidates, they vote for the candidate they are ideologically closer to. Uninformed voters are assumed to vote for the down-ballot co-partisan of their preferred presidential candidate. That is, if an uninformed voter votes for the Democratic presidential candidate, she votes for the Democratic down-ballot candidate, too. This assumption is grounded in the idea that uninformed voters are more likely to be low-propensity voters [Halberstam and Montagnes, 2015]; one explanation for this is that they may have higher costs of voting, including informational costs. Therefore, they are more likely to rely on partisan affiliation and (more easily accessible) information about the top-ballot candidate to determine their down-ballot vote choice.

Suppose for now that all citizens vote. Let π be the proportion of uninformed citizens that prefer the Democratic presidential candidate (and therefore will vote for the Democratic down-ballot candidate). Then, the Democratic down-ballot candidate wins if and only if

$$\frac{y_D + y_R}{2} > (1 - 2\pi) \frac{\delta}{1 - \delta}, \quad (1)$$

where y_D and y_R are the ideological positions of the Democratic and Republican down-ballot candidates.¹⁵

Halberstam and Montagnes posit that informed voters likely have more extreme ideological positions than uninformed voters (and document this with survey data). As such, informed voters likely act as strong partisans and are thus unaffected by valence realizations. However, uninformed voters, who likely have more moderate policy preferences, may choose to switch their vote or abstain entirely in the presence of a negative valence shock for their ideologically-preferred candidate (i.e., act as weak partisans or swing voters). If we endow presidential candidates in this model with some valence characteristic and assume that a negative valence shock for the Democratic presidential candidate can lead a left-leaning uninformed voter to either vote for the Republican presidential candidate or abstain entirely, then given that uninformed voters are straight-ticket voters, the presidential candidate's valence shock can affect the down-ballot Democrat, who loses voters through decreased turnout and/or vote-

¹⁵Note: this is equation (1) in Halberstam and Montagnes [2015].

switching.¹⁶

First, consider the vote-switching mechanism. Note that the right-hand side of condition (1) is decreasing in π , the share of uninformed voters that prefer the Democratic presidential candidate. If the Democratic presidential candidate experiences a negative valence shock, then some uninformed voters who are ideologically closer to the Democratic presidential candidate will now prefer her Republican opponent (vote-switching), decreasing π . As shown by Halberstam and Montagnes, for a given y_R , the probability the Democratic down-ballot candidate wins is $\min\{y_R - \frac{2\delta}{1-\delta}(1-2\pi), 1\}$, which is increasing in π . Therefore, as π decreases from the negative valence shock to the Democratic presidential candidate, the likelihood that the Democratic down-ballot candidate wins also decreases.

Next, consider the turnout mechanism. Suppose that in the face of a negative valence shock for the Democratic presidential candidate, a proportion $\gamma > 0$ of uninformed voters who are ideologically closer to the Democratic presidential candidate choose to abstain.¹⁷ Here, I assume no change in the proportion of uninformed voters who prefer the Democratic presidential candidate (i.e., π remains unchanged) in order to isolate the turnout mechanism: preferences over presidential candidates are unchanged, but some left-leaning uninformed voters simply opt to remain home. Then, the Democratic down-ballot candidate wins if and only if

$$\frac{y_D + y_R}{2} > \frac{\delta}{1-\delta}(1 - \pi(2 - \gamma)), \quad (2)$$

where the right-hand side is increasing in γ . For a given y_R , the probability the Democratic down-ballot candidate wins is thus $\min\{y_R - \frac{2\delta}{1-\delta}(1 - \pi(2 - \gamma)), 1\}$, which is decreasing in γ . Therefore, as γ increases (as it would through the turnout mechanism: $\gamma = 0$ without abstentions, yielding condition (1)), the likelihood that the Democratic down-ballot candidate wins decreases.

¹⁶To focus on the impact of presidential candidates' valence, I choose not to assign valence characteristics to down-ballot candidates, similar to the assumption that House candidates are of equal quality in Knight [2017]. Given that uninformed voters do not observe down-ballot candidates' policy positions, it is plausible they would not be aware of down-ballot candidates' quality anyway.

¹⁷Assume that uninformed voters who are ideologically closer to the Republican presidential candidate were all already voting, so their behavior is unchanged. If we relaxed this assumption and made voting costly, then this shock may induce some right-leaning uninformed voters previously planning to stay home to turn out, strengthening the effects.

From conditions (1) and (2), we can see how the Comey letter may have generated coattail effects for down-ballot Democrats in 2016. In particular, uninformed voters, who are more likely to be ideologically moderate [Halberstam and Montagnes, 2015], are assumed to be straight-ticket voters who align their down-ballot vote with their presidential one. Moderate voters respond to negative valence shocks to their ideologically-preferred presidential candidate by either voting for her opponent or abstaining entirely [McCrary, 2025], creating spillover effects for down-ballot Democrats through lost votes due to vote-switching and/or decreases in turnout.

3.1 Evidence from Survey Data

Survey data from the Cooperative Congressional Election Study administered in 2016 provides suggestive evidence of both the turnout and vote-switching mechanisms. Of the 10,307 respondents who 1) took the pre-election survey ahead of 10/28, 2) reported preferring Clinton in the pre-election survey, and 3) reported voting on Election Day (so that we can be sure they voted with knowledge of the Comey letter), 352 reported voting for Trump, and 130 reported voting for a third-party candidate. However, to interpret this as vote-switching due to the Comey letter alone, we must make the assumption that no other events or information between when they completed the pre-election survey and Election Day influenced their vote choice.¹⁸

Figure 3 documents daily averages of CCES respondents’ turnout intentions throughout the weeks leading up to Election Day in 2016. There is a decline in intended turnout following the date of the letter’s release (October 28th) – a sharper decline than we would have expected based on the trend in responses recorded prior to 10/28. Given that uninformed voters are more likely to be marginal voters, these turnout effects may be concentrated among straight-ticket voters, generating spillover effects for down-ballot Democrats through the turnout mechanism described above. I formally evaluate the impacts of the letter on election outcomes in Section 4. With the data available, there is no clear way to formally evaluate the vote-switching mechanism, but I estimate the letter’s effect on partisan turnout, providing formal evidence of the turnout mechanism.

¹⁸Limiting the sample to respondents who took the pre-election survey after October 7th, the date of the release of the Access Hollywood tape and the beginning of the WikiLeaks releases of John Podesta’s emails, the sample becomes 7,354 voters. Of those 7,354 respondents, 317 (resp., 109) reported voting for Trump (resp., a third-party candidate) on Election Day.

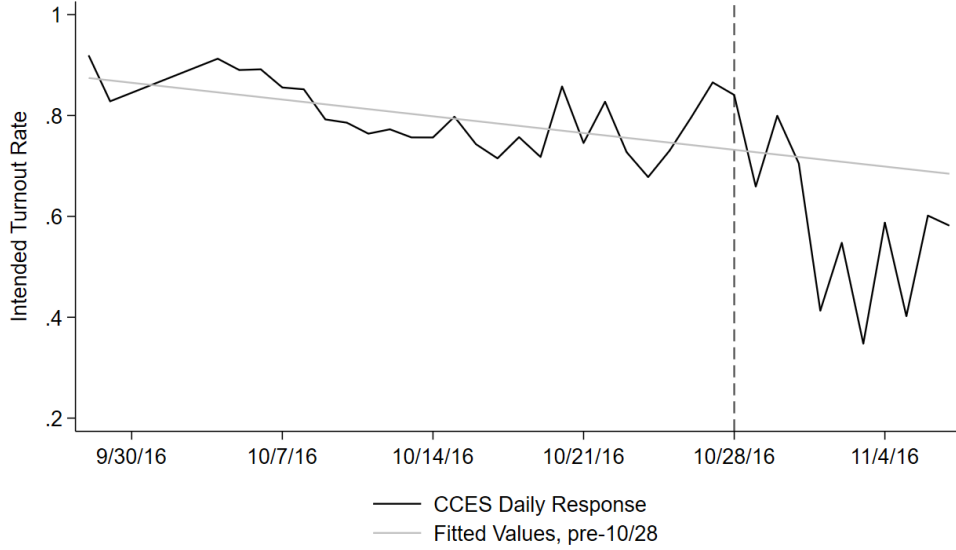


Figure 3: This figure shows the CCES’ daily average intended turnout rate relative to a trend line based on pre-10/28 responses only. The average is calculated using CCES’ survey weights so that it is nationally representative.

It should be noted that since I theorize that the letter’s effects were driven by changes in turnout or vote-switching by uninformed, low-propensity voters, the findings in this paper may not be generalizable to coattails generated by top-ballot candidates in midterm elections, where low-propensity voters are much less likely to turn out. At the same time, if a top-ballot candidate in a midterm year is popular enough to mobilize low-propensity voters, one could imagine positive spillovers for their down-ballot co-partisans. Further research is needed to understand the mechanisms which generate coattail effects in midterm elections.

4 Data & Methodology

In this section, I present the data and methodology used in my analysis. I make use of three complementary empirical strategies in this paper. The main design, which relies on cross-state variation in the availability of early voting, allows me to examine how the Comey letter impacted voter turnout and partisan presidential vote shares. In addition, I study the performance of down-ballot candidates to determine if the letter had spillover effects based on party affiliation. The second specification compares votes cast on Election Day vs. during the early voting period in the six states which report these totals separately (Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina). The third identification strategy

leverages a unique feature of North Carolina’s election data – daily partisan early voting turnout data – to compare pre-Comey letter early voters to post-Comey letter early voters. The latter two designs allow me to rule out confounding differences between early voting and non-early voting states, but at the cost of a smaller sample size.

4.1 Data

I obtain data on convenience voting laws and timing from the Elections & Voting Information Center for the 2012 election [Hicks, 2012] and Ballotpedia for the 2016 election [Ballotpedia, b] (Accessed: Sep. 2021) and use it to define treatment in the central difference-in-differences design: states with early voting windows which open before 10/28/16 are control states and those without early voting windows open before 10/28/16 are treatment states (this includes states which did not have early voting at all). Treatment is therefore defined by the *lack* of ability to vote early without an excuse before the letter’s release, as voters in these states would have been less constrained in their means to respond to the letter in the ballot box.

I collect county-level presidential, congressional, and gubernatorial election returns (vote totals for each candidate) from Dave Leip’s Atlas of Elections [Leip] and counts of 18+ population (i.e., voting-age population) for each county from the Survey of Epidemiology and End Results (SEER). I use these counts as voting-age population for calculating turnout rates. For the two complementary designs, I collect county-level vote returns and turnout reports from each state’s election administrator. I obtain county-level control variables from the U.S. Census Bureau’s American Community Survey 5-year estimates, the U.S. Bureau of Labor Statistics, and the U.S. Census Bureau’s Census of Population and Housing.

4.2 State-Level Variation in Convenience Voting

In the main empirical specification, I categorize two types of counties as treated: counties in states without no-excuse early voting and counties in states where early voting did not open until after October 28th, 2016. The treatment turns on with the Comey letter (only in 2016 and not in 2012). I use counties in states with early voting which opened before October 28th, 2016 as control units, as voters in these states would have been least able to respond to the information. Here, the variation of interest comes from the availability of no-excuse

early voting, something set independently by each state.¹⁹

The identifying assumption here is that, absent the Comey letter, treatment and control counties would have trended similarly in the outcomes of interest (turnout rate and partisan vote shares). This design also requires that any late-election information in 2012 did not affect political behavior.²⁰ I estimate

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ct}, \quad (3)$$

where T_c takes a value of 1 for counties in states without early voting and states where early voting did not open until after October 28th, 2016 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 otherwise.

Outcomes of interest y_{ct} for county c in year t include turnout rate, vote shares for Democratic and Republican presidential candidates, and average vote shares for Democratic and Republican state-level candidates. I calculate turnout rate as a proportion of voting-age population; presidential vote shares are proportions of total ballots cast for the president. To determine the average partisan vote shares for state-level candidates, I average together the Democratic (or Republican) vote shares in U.S. House, Senate, and gubernatorial elections.²¹ I weight regressions by county-year voting-age population, as is common in the literature [Charles and Stephens Jr., 2013, Brunner et al., 2011], and cluster at the state-level.

4.2.1 Support for Identification

One potential threat to identification with the comparison of early voting to non-early voting states is the possibility that these two sets of states may have been experiencing different political trends. If this were true, we might expect to see it directly reflected in those states' ballots. Perhaps one set of states had a separate reason to swing left or right in 2016: toss-up down-ballot races, ballot initiatives, etc. I collect data on states' 2012 and 2016 ballot characteristics in four broad categories: toss-up Senate or gubernatorial races, the number and type of ballot initiatives, third-party presidential candidates on the ballot, and straight-

¹⁹I exclude from my analysis any state which changed its convenience voting laws between 2012 and 2016 (Colorado, Massachusetts, Minnesota, and Michigan).

²⁰Identification requires that states' early voting laws do not change between reference elections and 2016, so that any results are attributable to the Comey letter and not changes in convenience voting opportunities. I opt to use only 2012 as a comparison year to keep the sample of states as large as possible, as well as to satisfy the requirement that reference elections not feature October surprise(s) near October 28th.

²¹I exclude unopposed elections, as vote shares would not reflect any spillover effects in these cases.

ticket voting.^{22,23} I also collect data on states’ “battleground status,” as states expected to have especially competitive presidential races may have drawn outsized campaign activity.²⁴ Results from a balance test using these variables are shown in Section A.1.1; I find that there is no relationship between treatment and any of these items on the ballot or battleground status.

Another way to test for differing political trends is to examine if campaign activity differed across treatment and control states. For example, if campaign spending were differentially high in treated states in 2012 or 2016, this would pose an issue for identification. I collect data on total campaign spending and campaign donations to candidates in 2012 and 2016 from the Federal Election Commission. As an additional measure of campaign activity, I use data from the Wesleyan Media Project on total campaign advertisements favoring Democratic or Republican candidates in 2012 and 2016. I aggregate these data to the state-week level, allowing me to look at trends throughout the election cycles in 2012 and 2016. I estimate separate two-way fixed effects models for 2012 and 2016 and implement a triple-difference strategy to compare the two years directly. There are no significant pretrends in (logged) campaign expenditures or donations, and differences in these measures cannot explain my results. In 2016, there were increased campaign ads in *non*-early voting states in the very early weeks of the election, but this did not differ significantly by party. See Section A.1.2 for results.

To add further support for the assumption of parallel trends, I turn to estimates of voters’ intentions throughout the election cycle. The Cooperative Congressional Election Study interviews voters on their turnout and vote intentions. According to survey documentation, samples are sufficiently large so that survey responses are roughly representative of each state. Therefore, I can compare voters’ intent to turn out and, conditional on intent to vote, intent to vote for the Democratic presidential and U.S. House candidates in treatment and control states. Again estimating separate two-way fixed effects models for 2012 and 2016 and a triple-difference strategy to compare the two elections, I find no evidence of differential pre-trends among treatment and control states in either year (see Section A.1.2).

Finally, as another source of high-frequency estimates of voters’ intentions in the weeks leading up to the election, I use data from Intrade and PredictIt on daily closing prices

²²Toss-up races are the most competitive races of the election cycle, where neither party is predicted to have a slight edge.

²³Straight-ticket voting allows a voter to vote for all of their chosen party’s candidates for each partisan office on the ballot with a single mark of their ballot.

²⁴I categorize all states deemed to have competitive presidential races by Cook Political Report as battleground states.

for the Democratic presidential candidate’s market in a particular state (representing daily predictions of the state-level Democrat vote share) for 2012 and 2016.²⁵ Using the same methods described above, I find no significant pretrends in the triple-differences specification (see Section A.1.2). I further address identification concerns with two complementary empirical specifications and several robustness checks, introduced below.

4.3 Comparisons of Early and Election Day Voters

While my main empirical strategy relies on comparing states with and without early voting, I am also able to compare early voters and Election Day voters *within* early voting states. Six early voting states (Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina) report votes cast on Election Day and during the early voting period separately. I leverage this disaggregated data to compare votes cast by early voters and Election Day voters in 2012 and 2016, considering Election Day voters as “treated,” as they voted after the letter’s release in 2016. It should be noted that early voting extended past 10/28 in each of these states, and so some of the early voters (“controls”) would have had knowledge of the letter when casting their vote.²⁶ However, I still expect to observe the Comey effect under the assumption that sufficiently many early voters cast their ballot ahead of 10/28 in 2016, possessing no knowledge of the letter when they did so.

For this reason, I expect to observe larger effects in states that had more (vs. less) days of early voting available ahead of the letter’s release. In states with more days, vote shares from early voting periods should be more representative of voters’ preferences without knowledge of the letter, as it is much more likely that many of them cast their ballot ahead of the letter’s release (unlike, for example, Maryland, where early voting began on 10/27). Figure 4, which plots the relationship between the number of early voting days ahead of the letter’s release and early voting turnout rate, provides suggestive evidence in favor of this argument. States with more early days of voting have more voters cast their ballot early. While I cannot be sure that a significant portion of these votes were cast prior to the letter’s release, I argue this is a more plausible explanation of the relationship than the alternative: states with more days of pre-10/28 early voting experiencing a differentially large concentration of early voting in the final days ahead of Election Day.

²⁵This data is only available for twenty-one states in 2016, but there is a good mix of treatment and control states represented: 12 control and 9 treated.

²⁶In this way, this strategy mirrors how early voting states are considered “partially shielded” from the Comey letter and thus used as controls in my first specification. Here, the analogous group is the early voters from these states.

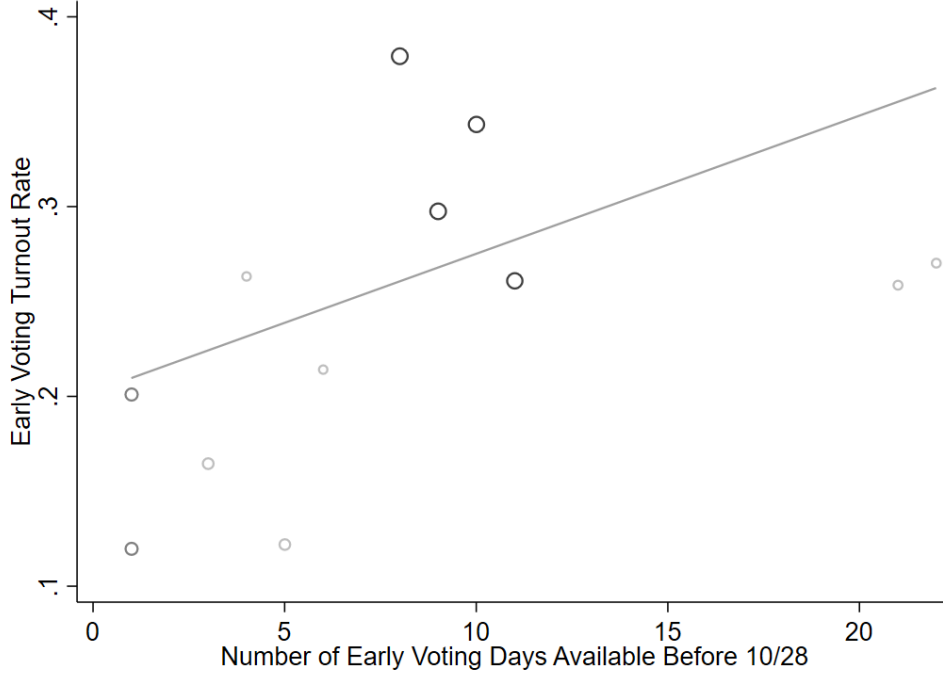


Figure 4: This figure shows a binned scatterplot (weighted by voting-age population) of the number days of early voting available ahead of 10/28 and early voting turnout rate (out of voting-age population) for states used to estimate Table 3. Markers are weighted by voting-age population.

Motivated by the relationship described in Figure 4, I consider a difference-in-differences specification with continuous treatment, leveraging the difference between counties in states with many vs. few days of early voting prior to 10/28. In addition to the traditional parallel trends assumption, this specification requires a further “strong parallel trends” assumption [Callaway et al., 2024]. In particular, I must assume that counties with many days of early voting prior to 10/28 would have experienced similar effects to counties with few days of early voting prior to 10/28 had they received a “lower dose” (fewer early voting days ahead of the letter’s release) instead. I estimate

$$y_{pcst} = \alpha + \delta T_p + \gamma P_t + \eta F_c + \beta(D_{st} \times T_p \times P_t) + \epsilon_{pcst}, \quad (4)$$

where T_p takes a value of 1 for votes cast on Election Day and 0 for votes cast in early voting periods, and P_t takes a value of 1 in the 2016 election and 0 otherwise. I include county fixed effects, F_c , to account for any time-invariant differences across counties which affect election

outcomes. The “dose,” D_{st} , is the number of days early voting was available prior to 10/28. This is determined independently by each state and only enters the specification in 2016. As such, I cluster at the state-level. Due to the very small number of clusters (6), I use a wild cluster bootstrap with the Webb six-point distribution [Roodman et al., 2019, Cameron and Miller, 2015]. As in my first specification, I weight regressions by county-year voting-age population. Outcomes of interest are Democratic and Republican presidential vote shares.

4.4 Comparisons of Pre- and Post-Comey Letter Early Voters

In the first two empirical designs, control voters are only partially shielded from the Comey letter: early voting periods extended past October 28th, so early voters (or voters in early voting states) are simply considered “more constrained” in their ability to respond to the letter, as a significant amount of them would have already voted. To my knowledge, no state reports votes received by each candidate at a finer level of disaggregation than the one leveraged in Section 4.3. However, North Carolina reports daily counts of early votes cast by party registration. I use this disaggregated data on partisan early voting turnout to compare pre-October 28th early voters to post-October 28th early voters, as only the latter group was able to incorporate information from Comey’s letter into their vote in 2016.

The unique system of reporting early voting turnout in North Carolina grants me the ability to make within-state comparisons and compare early voters to other early voters. That is, this method rules out confounding cross-state and cross-voting method differences. Additionally, studying turnout by party registration provides insight into the mechanisms discussed in Section 3. Thus far, I have been unable to distinguish between turnout and vote-switching: if I find no effect on turnout rates in the main specification, this could be caused by either 1) offsetting increases in turnout by moderate right-leaning voters and decreases in turnout by moderate left-leaning voters, or 2) vote-switching among moderate left-leaning voters, while turnout remains unchanged. However, a decrease in turnout by registered Democrats would indicate changes along the extensive, rather than intensive, margin. Although they come at the cost of examining only one state, these are valuable contributions. In the appendix, I provide further evidence of the turnout mechanism using weekly partisan early turnout data reported by Nevada in a similar design (see Table A10).

The daily structure of the North Carolina data also allows me to estimate a two-way fixed effects event study. I estimate

$$y_{cdt} = \delta_d + \delta_t + \sum_{k=-19}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_t + \sum_{k=-11}^{-3} \gamma_k \mathbf{1}\{d = k\} \times treat_t + X_{ct} + \epsilon_{cdt}, \quad (5)$$

where c indexes county, d indexes days relative to Election Day, and t indicates year. Note that $treat_t$ is a dummy variable equaling 1 in 2016 and 0 in 2012, and X_{ct} is a vector containing number of early voting sites, total number of early voting hours, number of evening early voting hours, number of Saturday early voting hours, and number of Sunday early voting hours. The letter was released eleven days ahead of Election Day, so I omit the day before. As in the first two empirical strategies, I weight by county-year voting-age population. Outcomes of interest are turnout by party registrants (i.e., voters registered as Democrats, etc.) as a share of total early votes.

It should be noted that ahead of the 2016 election, a North Carolina election reform law (passed in 2013) was partially stricken down by a federal appellate court, leaving counties with the ability to change hours and locations for early in-person voting. The majority of counties expanded early in-person hours, but 17 counties chose to restrict them. Using a border-county analysis, [Walker et al. \[2019\]](#) find no evidence of uniform effects on turnout from changes to early in-person total hours, evening hours, or Saturday hours, nor from changes to number of early in-person voting sites. They do find effects for some subgroups of voters in response to changes in the availability of Sunday early in-person voting, so I omit Sunday voting from the above analysis. I use data from [Walker et al. \[2019\]](#) to control for county-level early voting characteristics (number of sites, hours, etc.) and cluster at the county-level to account for these changes.

All counties continued to offer early voting each weekday beginning 19 days ahead of Election Day. While the director of the North Carolina Republican Party encouraged county elections board members to limit the availability of early voting, many of those same officials did not support plans to limit early voting hours [[Campbell, 2016](#)]. Given the push from a top Republican official to limit early voting, however, we might expect that counter-mobilization efforts led to higher turnout from Democrats in the first few days of early voting. With these caveats, there is no reason to expect that the changes to early in-person voting would have had differential effects for Democrat versus Republican voters beginning on the 9th day of early voting, which is what I test for with equation (5).

5 Results

Table 2 shows results from the main empirical design. First, note that there is no change in turnout. Since there is not a significant change on the extensive margin, I look for changes on the intensive margin by examining the effect of the Comey letter on presidential vote shares (total ballots cast for a candidate, divided by total presidential ballots cast). The results in Table 2 show an increase in Republican presidential vote share of 3.2 percentage points, in combination with a decrease in Democratic presidential vote share of 2 percentage points.

Table 2 also displays an increase in average Republican down-ballot vote share of 2.9 percentage points and a complementary decrease in average Democratic down-ballot vote share of 3.2 percentage points. As mentioned above, I drop uncontested races from my analysis, explaining the difference in the number of observations across columns. In Section 6, I discuss interpretation of these magnitudes.

Table 2: Voters most able to respond to the Comey letter were more likely to vote for Republican candidates in presidential and down-ballot races.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	0.009 (0.006)	0.008 (0.006)	-0.02* (0.011)	-0.024 (0.015)	0.032*** (0.01)	0.035** (0.015)	-0.032** (0.014)	-0.034* (0.017)	0.029* (0.017)	0.03 (0.018)
N	5782		5782		5782		5400		5400	
Mean	0.56		0.347		0.618		0.37		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-level difference-in-differences specification, weighted by voting-age population: $y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ct}$, where T_c takes a value of 1 in counties without early voting open before 10/28/16 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor's degree or higher, and share white without a college degree. I calculate turnout rate as a proportion of voting-age population; presidential vote shares are proportions of total presidential votes cast. I calculate U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in the respective races before averaging the three together to calculate the down-ballot vote share. I drop observations with uncontested races, leading to the different number of observations for the down-ballot outcomes.

Results are robust to the inclusion of time-varying county economic and demographic controls including median income, unemployment rate, population density, median age, number

of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor’s degree or higher, and share white without a college degree (right-hand columns under each outcome). Additionally, results are robust to the inclusion of a term interacting 2016 with share white without a college degree, given the large gap in 2016 among white voters by education [Doherty et al., 2018]: see Section A.2.1.

Note that counties in Florida and Oklahoma may be a superior “treated” group in this specification, as both Florida and Oklahoma had no-excuse early voting periods that did not open until after 10/28/16. In a specification using only Florida and Oklahoma counties as treated units, both treated and control voters live in states with early voting: the difference is just in when early voting is offered. Results are robust to such a refinement of the design (see Section A.2.2).

Identification also requires that we assume that any late-election information in 2012 did not affect turnout or partisan vote shares. Some may view Hurricane Sandy, which made landfall in the U.S. on October 29th, 2016, as late-election information which benefited President Obama (since his handling of the storm was positively regarded by both parties). See Section A.2.3 for a robustness check; results are robust to leaving out counties in Connecticut, New Jersey, and New York, which had the largest populations in storm surge zones during Hurricane Sandy [Center for International Earth Science Information Network, 2012].

It is possible that this approach picks up larger dynamics within the election, rather than voters’ response to the singular shock of the Comey letter. To ensure this is not the case, I conduct placebo tests using three alternative dates to define treatment: 9/26/16 (the first presidential debate), 10/7/16 (Access Hollywood tape release and start of Wikileaks Podesta email releases), and 10/19/16 (the final presidential debate). The results in Table 2 cannot be replicated when defining treatment around any of these dates (results in Section A.2.4). This suggests that the difference-in-differences design identifies the effect of the Comey letter.

A similar concern is that this approach captures differential trends across treatment and control states over a number of elections: perhaps non-early voting states were becoming more right-leaning relative to early voting states. To address this, I conduct two additional placebo tests. Keeping the definition of treatment the same, I re-estimate equation (3) comparing 1) the 2012 and 2008 elections (with 2012 as the “treated” election) and 2) the 2014 and 2018 elections (with 2018 as the “treated” election). I find no effects of treatment in either of these specifications (shown in Section A.2.5), lending support to the idea that I am estimating the effect of the Comey letter itself and not broader political trends in treatment

and control states.

Finally, in Section A.2.6, I estimate results from Table 2 separately for presidential battleground and non-battleground states. There is no effect of treatment on states’ battleground status (as shown in Figure A1), but it is plausible that the letter’s effect differed in these two sets of states due to varying levels of campaign activity. However, down-ballot effects of the letter were of importance in both sets of states: nearly one-half of battleground U.S. House races and one-third of battleground Senate races in 2016 took place in non-presidential battleground states. I find that down-ballot effects on Democratic and Republican vote shares and the effect on Republican presidential vote share do not significantly differ between the two groups.

Table 3 shows results from the design comparing partisan vote shares from early voting periods versus Election Day. Due to the small number of clusters (six states), I implement a wild cluster bootstrap using the Webb six-point distribution, estimated in Stata with the “boottest” command [Roodman et al., 2019]. Note that standard errors are not estimated as normality is not assumed here; instead, I report p -values in parentheses and 95 percent confidence intervals in square brackets. Results indicate that knowledge of the letter led to a significant increase (resp., decrease) in Republican (resp., Democratic) presidential vote share.

In particular, I find that that a county with 10.17 days of early voting ahead of the letter’s release (the average number of days of early voting prior to 10/28 across all states with no-excuse early voting in 2016) saw a decrease of 3.05 percentage points in Democratic presidential vote share and an increase of 2.03 percentage points in Republican presidential vote share in Election Day voters relative to early voters. The decrease in Democratic presidential vote share is robust to a non-continuous specification (see Appendix Table A9).

These results are quite similar to those obtained by comparing states with and without early voting (reported in Table 2), where I find exposure to the letter decreased Democratic presidential vote share by 2 p.p. and increased Republican presidential vote share by 3.2 p.p. These two empirical strategies leverage similar, but separate, sources of variation and produce comparable results. Both sets of estimates also may represent underestimates of the Comey letter’s effect, as both strategies include some voters who cast their ballot after the letter’s release as controls. In the first specification, these are voters in early voting states who either voted during the early voting period but after 10/28 or on Election Day. In the second, these are early voters who voted after 10/28.

Table 3: Voters most able to respond to the Comey letter were less likely to vote for the Democratic presidential candidate and more likely to vote for the Republican presidential candidate.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share	
# of Days EV	-0.003	-0.003	0.002	0.002
×Treat×Post	(0.000)	(0.000)	(0.009)	(0.003)
	[-0.0039, -0.0014]	[-0.0039, -0.0014]	[0.0005, 0.0029]	[0.0006, 0.0028]
<i>N</i>	2072		2072	
Mean	0.377		0.595	
Controls	✓		✓	

Wild cluster bootstrap p -values are listed in parentheses, and wild cluster bootstrap 95 percent confidence intervals are given in square brackets.

Notes: This table shows results from a county-period level continuous difference-in-differences specification, weighted by voting-age population and clustered at the state level: $y_{pcst} = \alpha + \delta T_p + \gamma P_t + \eta F_c + \beta(D_{st} \times T_p \times P_t) + \epsilon_{pcst}$, where T_p takes a value of 1 in the Election Day voting period and 0 in the early voting period, and P_t takes a value of 1 in 2016 and 0 in 2012. County fixed effects are given by F_c , and D_{st} gives the number of days of early voting available before October 28 in state s in year t . I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor's degree or higher, and share white without a college degree. Presidential vote shares are proportions of total presidential votes cast.

Figure 5 displays results from two-way fixed effects event studies examining daily partisan early voting turnout shares in North Carolina. Recall that several counties made changes to early in-person voting hours and number of sites in 2016 (for this reason, I cluster at the county-level and control for numerous measures of early voting availability: number of early voting sites, total number of early voting hours, number of evening early voting hours, number of Saturday early voting hours, and number of Sunday early voting hours). As discussed in Section 4.4, counter-mobilization efforts may have led to increased turnout by Democrats when early voting first opened. However, there is no reason to expect that these changes would have led to differential turnout rates by party around the date of the letter's release.

I find a 2.3 p.p. decrease in registered Democrats' early turnout share and a corresponding 2.3 p.p. increase in registered Republicans' early turnout share (as a proportion of all voters,

including those not registered as a Republican or Democrat) following the letter’s release.²⁷ Note that we observe the strongest effects immediately after the letter’s release, but there are no offsetting effects later on. That is, it is not the case that registered Democrats were deterred initially but made up for this drop in turnout a few days later.

These findings are consistent with the turnout mechanism discussed in Section 3; i.e., they suggest that the letter caused a decline in turnout among left-leaning voters who likely would have supported Clinton (and her down-ballot co-partisans), generating coattail effects. In the appendix, I include a table of the results from a two-way fixed effects specification using this data and demonstrate that results are robust to the omission of the first three days, where mobilization efforts appear to have increased Democrat turnout. In addition, I provide further evidence of the turnout mechanism by leveraging weekly partisan early turnout data in Nevada in a similar design, finding a decrease in turnout by registered Democrats after 10/28 in 2016 (see Table A10).

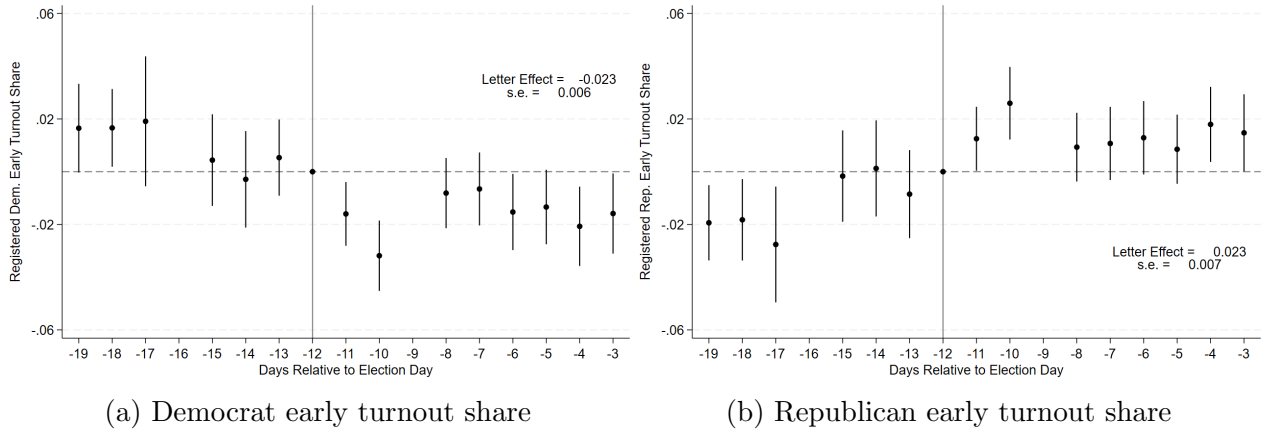


Figure 5: After the Comey letter, the daily share of ballots cast by Democrats (resp., Republicans) in North Carolina decreased (resp., increased).

Notes: This figure displays results from a two-way fixed effects event study specification, weighted by voting-age population: $y_{c dt} = \delta_d + \delta_t + \sum_{k=-19}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_t + \sum_{k=-11}^{-3} \gamma_k \mathbf{1}\{d = k\} \times treat_t + X_{ct} + \epsilon_{c dt}$, where c indexes county, d indexes days relative to Election Day, and t indexes year. Note that $treat_t$ is a dummy variable equaling 1 in 2016 and 0 in 2012, and X_{ct} is a vector of county-level early voting characteristics. The letter was released 11 days ahead of Election Day, so I omit the 12th day. Standard errors are clustered at the county level; the graph depicts 90% confidence intervals. I omit Sunday voting, explaining the gaps at -16 and -9 days relative to Election Day.

²⁷The letter was dominating news headlines by 2:00 PM eastern on October 28th [Silver, 2017], so that several hours of early voting were left on the day of the letter’s release. For example, Wake County, North Carolina’s most populous county, had early voting sites open until 7:00 PM that day (<http://msweb03.co.wake.nc.us/bordelec/downloads/14earlyvoting/info/EarlyVotingSitesWithTurnout.pdf>).

6 Interpreting Magnitudes: An Instrumented Difference-in-Differences Approach

In Table 2, I find that exposure to the Comey letter led to significant decreases in down-ballot Democratic vote shares (and increases in their Republican counterparts). These findings are suggestive of presidential coattail effects: the popularity of a presidential candidate affected the performance of her party’s candidates in concurrent races. Scholars have long sought to estimate presidential coattail effects. However, this is often a difficult task: rarely can the popularity of a presidential candidate be separated from the general attitudes toward her political party.

In order to estimate presidential coattail effects in my setting, I turn to an instrumented difference-in-differences (DDIV) model [Hudson et al., 2017, Duflo, 2001]. This model scales the difference-in-differences effect of the Comey letter on down-ballot partisan vote shares by its effect on partisan presidential vote shares. In other words, it allows me to obtain the effect of a one percentage point increase in partisan presidential vote share, rather than the effect of exposure to the Comey letter itself. The DDIV coefficient β is estimated via the following instrumental variables system:

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta \widehat{Y}_{ct} + \epsilon_{ct} \quad (6)$$

$$Y_{ct} = \theta + \rho T_c + \pi P_t + \eta(T_c \times P_t) + \mu_{ct}, \quad (7)$$

where T_c and P_t are defined as in equation (3). Here, exposure to the Comey letter instruments for Y_{ct} , the Republican presidential vote share in county c in year t .^{28,29} In the second-stage equation, outcomes of interest y_{ct} include average vote shares for Republican and Democratic state-level candidates. I weight by voting-age population in both estimations.

Note that the first-stage parameter, η from equation (7), is equivalent to the coefficient estimated by equation (3) when regressing on presidential vote shares.³⁰ The DDIV coef-

²⁸I instrument for Republican presidential vote share rather than Democratic for two reasons: a stronger first-stage and ease of interpretation (exposure to the Comey letter increased the Republican – not Democratic – presidential vote share).

²⁹In the formal treatment of DDIV provided in Hudson et al. [2017], Y_{ct} is assumed to be “a discretely-valued and positively-valued treatment.” However, the authors note that it is straightforward to extend to a setting with continuous treatment – such as a presidential vote share.

³⁰For the IV exercise, I drop all counties which only had one major party run in a state-level race. This results in a sample of 5400, rather than 5782, observations, and means that η from equation (7) and β from equation (3) differ slightly.

ficient β can be written as the ratio of the reduced-form and first-stage parameters. The reduced-form parameter can be found by estimating:

$$y_{ct} = \nu + \tau T_c + \xi P_t + \phi(T_c \times P_t) + \omega_{ct}, \quad (8)$$

i.e. equation (3) when regressing on state-level candidates' vote shares. As discussed in Hudson et al. [2017], the DDIV design requires several assumptions. Just as in the difference-in-differences model estimated in equation (3), the assumption of parallel trends must hold. Similar to traditional instrumental variables models, monotonicity and the exclusion restriction must hold. In my context, this requires assuming 1) exposure to the Comey letter did not cause an increase in Democratic presidential vote share for some subpopulations and 2) knowledge of the Comey letter did not affect partisan down-ballot vote shares except through its effect on the popularity of each party's presidential candidates. Table 4 presents the results from equations (6) and (7) where the outcomes of interest are down-ballot partisan vote shares (averaged over U.S. House, Senate, and gubernatorial elections).

Table 4: An increase in the Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) down-ballot vote shares.

	(1)	(2)
Rep. Down-Ballot Vote Share	0.787** (0.334)	0.786** (0.325)
Dem. Down-Ballot Vote Share	-0.872*** (0.311)	-0.88*** (0.308)
First-Stage (Rep. Pres. Vote Share)	0.037*** (0.014)	0.038** (0.016)
N	5400	5400
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from an instrumental variables system. In the first-stage, I estimate $Y_{ct} = \theta + \rho T_c + \pi P_t + \eta(T_c \times P_t) + \mu_{ct}$, where T_c takes a value of 1 in counties without early voting open before 10/28/16 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. Here, Y_{ct} is the Republican presidential vote share (as a proportion of total presidential votes cast) in county c in year t . Then, I estimate $y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta \widehat{Y}_{ct} + \epsilon_{ct}$, where y_{ct} is the average of Republican or Democratic U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in their respective races. All specifications are weighted by voting-age population. In column (2), I include time-varying controls: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor's degree or higher, and share white without a college degree. I drop observations where only one major party competed in all relevant races.

Results indicate that a one percentage point increase in the Republican presidential vote

share leads to a 0.79 percentage point increase in the down-ballot Republican vote share.³¹

7 Conclusion

This paper is the first to make use of an October surprise to estimate presidential coattail effects: using an instrumented difference-in-differences design, I find that a one percentage point increase in Republican presidential vote share leads to a 0.79 percentage point increase in Republican down-ballot vote share. Results in this paper indicate that James Comey’s letter to Congress, released just eleven days ahead of the 2016 presidential election, increased both the presidential and down-ballot vote shares of Republican candidates. Using cross-state variation in the availability of early voting, I find that exposure to the letter led to an increase in the Republican presidential vote share of 3.2 percentage points – larger than the eight closest state margins that election.

Complementary empirical designs suggest that the effect of the letter is robust to limiting to both within-state and early voting state-only comparisons. In a specification comparing partisan vote shares among early voters and Election Day voters in early voting states, I find a differential increase in Republican presidential vote share in counties with more early voting days (vs. less) ahead of the letter’s release. I also provide evidence on the mechanisms of coattail effects: when comparing pre-10/28 early voters to post-10/28 early voters, I find that exposure to the letter led to a decrease in registered Democrats’ early turnout share of 2.3 percentage points in North Carolina. This is consistent with the theory that negative information regarding a presidential candidate can demobilize low-propensity, uninformed voters who likely would have voted for her down-ballot co-partisans had they turned out.

The findings of this paper indicate that the popularity or electoral strength of a party’s presidential candidate has meaningful downstream effects. These spillovers have important policy implications: namely, coattail effects increase the likelihood of unified, rather than divided, government. In leading to more occurrences of unified government, coattail effects may impact legislative efficacy and the types of policies passed. Coattail effects raise numerous strategic considerations for down-ballot political candidates, too: for example, how closely to align with prominent co-partisan candidates or even when to run for office. Finally, the finding that late-election information regarding a presidential candidate has down-ballot effects suggests that there are strong incentives for the strategic release of information during presidential campaigns beyond first-order effects on the presidential race.

³¹Note that much of the coattails literature is dedicated to estimating the extent of spillovers between presidential and U.S. House races. In Table A11, I re-estimate equations (6) and (7) using partisan U.S. House vote shares as the sole outcomes of interest.

A Appendix

A.1 Support for Identification

A.1.1 Balance Test

In equation (3), I leverage cross-state variation in preexisting convenience voting laws at the time of the Comey letter. Counties in states with early voting open prior to 10/28 are controls, and counties in states 1) with early voting open only *after* 10/28 and 2) with no early voting at all are treated. The identifying assumption is the parallel trends assumption: absent the Comey letter, treatment and control counties should have trended similarly in the outcomes of interest. However, one might be concerned that states with and without early voting might have been on different political trends between 2012 and 2016, violating this assumption.

I conduct a balance test in order to address this concern. I examine whether treatment is related to items on states' ballots whose presence may reflect varying political trends or a differential reason to turn out or to vote for a specific partisan candidate in that election year. I focus on four broad categories: number (and type) of ballot initiatives, toss-up Senate or gubernatorial races, the availability of straight-ticket voting, and the presence of a third-party presidential candidate. In addition, I consider states' battleground status, i.e. whether the presidential race was considered a competitive one in that state.

Ballot initiatives have been shown to increase voter turnout [Tolbert et al., 2001]; for this reason, I test whether the number of ballot initiatives is related to treatment. Different categories of ballot measures can serve to mobilize specific groups of voters [Garretson, 2014], potentially leading to spillover effects on other election outcomes. For example, Gardner et al. [2025] find that the presence of abortion-related ballot measures boosted Democrats' performance in congressional races in the 2022 midterm elections. I collect data from Ballotpedia on a few major categories of ballot measures (taxes, healthcare, labor, social issues) to test for a relationship between the presence of different ballot initiative categories and treatment.^{32,33}

³²In the 2012 presidential election, four states had ballot initiatives aimed at restricting the Affordable Care Act. I put these ballot measures in their own category separate from other healthcare-related initiatives.

³³The "social issues" category includes measures on gun rights, same-sex marriage, abortion, the death penalty, and marijuana legalization.

Especially competitive races (either presidential or state-level) might increase voter turnout, amplify partisan campaigning efforts, etc. I collect data on which states 1) were considered battleground states (i.e., those with competitive presidential elections) or 2) had toss-up Senate and gubernatorial races in 2012 and 2016, as defined by the Cook Political Report. Third-party presidential candidates were particularly important in the 2016 election, so I also tested whether the presence of third-party presidential candidates on the ballot can be predicted by treatment.

Since I leverage exposure to the Comey letter to estimate coattail effects, I am especially concerned with voters' access to straight-ticket voting, which should theoretically increase the strength of such effects (as it makes voting for a single party up and down the ballot particularly easy). In fact, several states did away with straight-ticket voting in the years preceding 2016. I use information from the National Conference of State Legislatures to code each state's straight-ticket status in 2012 and 2016.

I re-estimate equation (3) and place indicator variables on the left-hand side for each of these variables (or for number of ballot measures, number of ballot measures divided by ten). Results are shown in Figure A1. There is no statistically significant relationship between treatment and any of the covariates, supporting the parallel trends assumption.

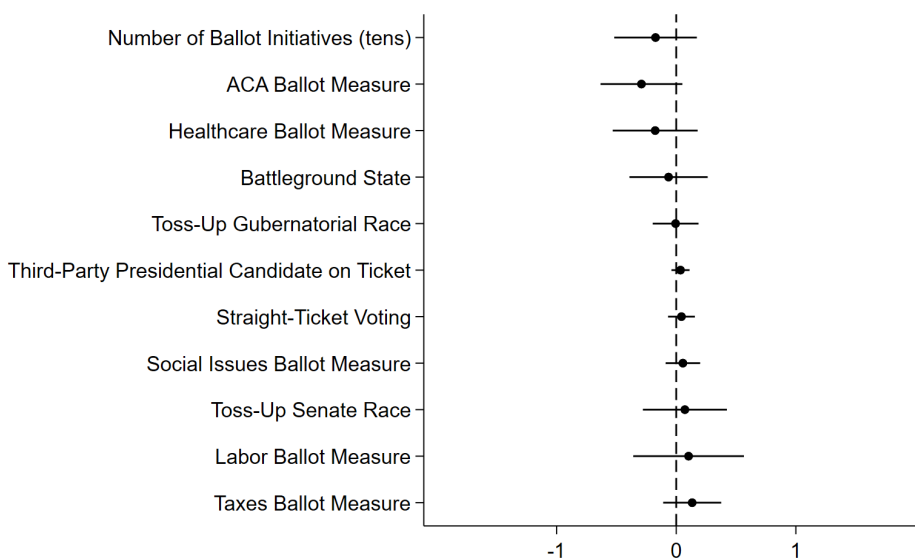


Figure A1: There is no effect of treatment on states' ballots or battleground status. That is, early voting and non-early voting states did not appear to be on different political trends between 2012-2016.

Notes: This figure displays results from variations of equation (3) which place ballot characteristics and battleground status on the left-hand side.

A.1.2 Comparisons of Treatment and Control States with High-Frequency Data

Identification in equation (3) requires that treatment and control states were trending similarly both pre and post-10/28 in 2012 and prior to 10/28 in 2016. Several high-frequency datasets allow for various tests of this assumption. In the following section, I use high-frequency data on campaign expenditures, campaign donations, television campaign advertisements, predicted vote shares of the presidential candidate from prediction markets, and survey data on voters' intentions to compare treatment and control states throughout the weeks leading up to Election Day in 2012 and 2016.

I estimate variations of two equations. First, I estimate a two-way fixed effects event study:

$$y_{sw} = \delta_s + \delta_w + \sum_{k < -3} \mu_k \mathbf{1}\{w = k\} \times treat_s + \sum_{k \geq -2} \gamma_k \mathbf{1}\{w = k\} \times treat_s + \epsilon_{sw}, \quad (9)$$

where s indexes state and w indexes weeks relative to Election Day. Here, $treat_s$ is a dummy variable equal to 1 in states without early voting available ahead of 10/28/16 and 0 otherwise. The Comey letter was released eleven days ahead of Election Day, so the third week prior to Election Day is omitted. I adapt the above specification to the daily level where relevant, omitting the twelfth day ahead of Election Day. I estimate equation (9) separately for 2012 and 2016.

Second, I estimate the following triple-difference specification:

$$y_{swt} = \delta_{st} + \delta_{sw} + \delta_{wt} + \sum_{k < -3} \mu_k \mathbf{1}\{w = k\} \times treat_{st} + \sum_{k \geq -2} \gamma_k \mathbf{1}\{w = k\} \times treat_{st} + \epsilon_{swt}, \quad (10)$$

where s indexes state, w indexes weeks relative to Election Day, and t indexes year. Here, $treat_{st}$ is a dummy variable equal to 1 in states without early voting available ahead of 10/28/16 (only turning on in 2016) and 0 otherwise. Again, I omit the week prior to the Comey letter's release and adapt to the daily level where relevant.

I first consider measurements of campaign activity in the form of campaign expenditures and campaign donations using data from the Federal Election Commission. I collect data on total expenditures from principal campaign committees of Republican and Democratic candidates and total donations made to Republican and Democratic candidates. I aggregate these data (available daily at the ZIP-level) to the state-week level, as such a fine level of measurement may introduce quite a bit of noise. I estimate equations (9) and (10), where the outcomes of interest are logged total campaign expenditures and logged total campaign

donations.³⁴ Results are shown in Figures A2 and A3. While estimates are noisy, there are no significant pretrends in either year.

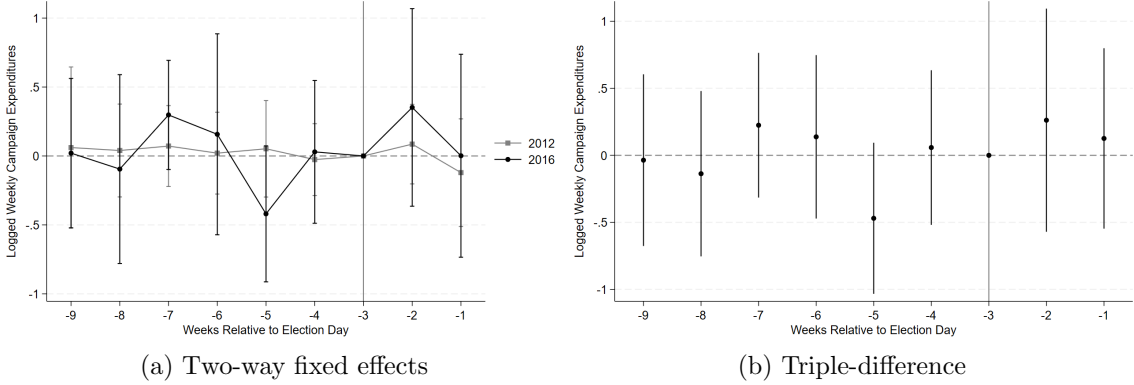


Figure A2: Logged weekly campaign expenditures did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure displays results from two event study specifications. In Panel A, two-way fixed effects models are estimated separately in 2012 and 2016 and displayed together. In Panel B, a triple-difference specification is estimated. Panel A shows results from estimating $y_{sw} = \delta_s + \delta_w + \sum_{k=-9}^{-4} \mu_k \mathbf{1}\{w = k\} \times treat_s + \sum_{k=-2}^{-1} \gamma_k \mathbf{1}\{w = k\} \times treat_s + \epsilon_{sw}$, where s indexes state and w indexes weeks relative to Election Day. Note that $treat_s$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 and 0 otherwise. Panel B shows results from estimating $y_{swt} = \delta_{st} + \delta_{sw} + \delta_{wt} + \sum_{k=-9}^{-4} \mu_k \mathbf{1}\{w = k\} \times treat_{st} + \sum_{k=-2}^{-1} \gamma_k \mathbf{1}\{w = k\} \times treat_{st} + \epsilon_{swt}$, where s indexes state, w indexes weeks relative to Election Day, and t indexes year. Note that $treat_{st}$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 (only turning on in 2016) and 0 otherwise. Each specification is weighted by voting-age population with standard errors clustered at the state level; 95% confidence intervals are displayed.

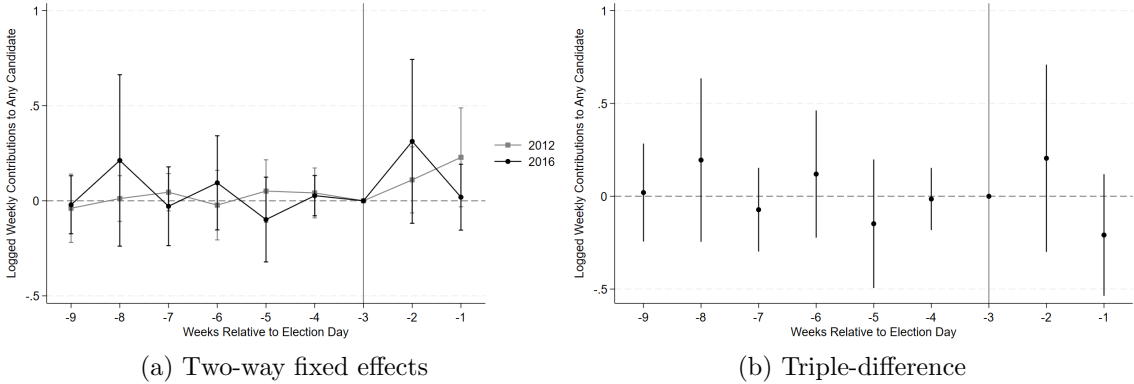


Figure A3: Logged weekly campaign donations did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure re-estimates equations used in Figure A2 using logged total campaign donations as the outcome variable. I weight by voting-age population and cluster at the state level; 95% confidence intervals are displayed.

³⁴I use $\ln(y + 1.1)$ to account for any zeros in the state-week panel.

Next, I turn to daily television campaign advertisement data from the Wesleyan Media Project. I collect data on total TV campaign ads from Democratic and Republican candidates. This data is provided at the Designated Media Market (DMA) level, which I cross-walk to the county level as DMAs can cross state borders. I then aggregate the data to the state-week level and estimate equations (9) and (10), where the outcome of interest is logged total TV campaign ads.³⁵ Results are shown in Figure A4. There were no significant pretrends in 2012. In 2016, there was increased focus on *non*-early voting states in the very early weeks of the election, but this did not differ significantly by party (see Figure A5).

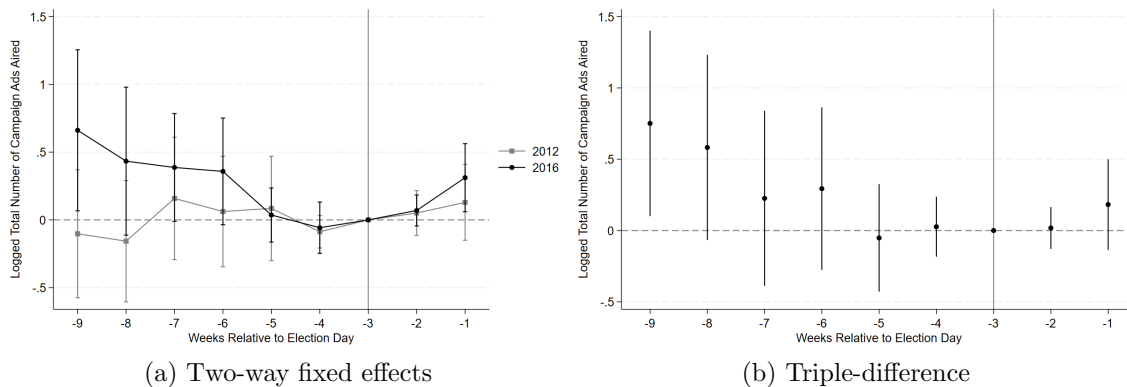


Figure A4: Logged weekly television campaign advertisements did not trend differently in early voting and non-early voting states in 2012. In 2016, there was increased focus on *non*-early voting states in the very early weeks of the election, but this did not differ significantly by party.

Notes: This figure re-estimates equations used in Figure A2 using logged number of television campaign advertisements as the outcome variable. I weight by voting-age population and cluster at the state level; 95% confidence intervals are displayed.

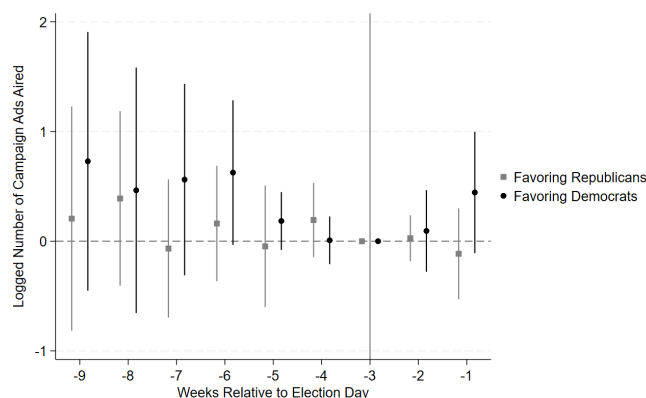


Figure A5: Triple-difference coefficients for logged television campaign advertisements did not differ significantly by party.

Notes: This figure displays the coefficients from the triple-difference event studies for two different outcomes: logged campaign ads favoring Republicans and logged campaign ads favoring Democrats.

³⁵ Again, I use $\ln(y + 1.1)$ to account for any zeros.

I next consider daily closing prices for the Democratic presidential candidate’s market in each state (representing daily predictions of the state-level Democrat vote share). This data comes from Intrade for 2012 [Imai, 2017] and PredictIt for 2016.³⁶ This data is only available for twenty-one states in 2016, but there is a good mix of treatment and control states represented: 12 control and 9 treated.³⁷ I estimate equations (9) and (10), where the outcome of interest is closing price for the Democratic presidential candidate’s market (i.e., predicted Democrat presidential vote share). Results are shown in Figure A6. There are no significant pretrends in the triple-difference specification. Interestingly, there is an upward trend in predicted Democrat presidential vote share for treated states after the letter’s release (even though I find these groups saw a decrease in Democrat presidential vote share using election outcomes). This is likely due to the composition of states for which this data is available.

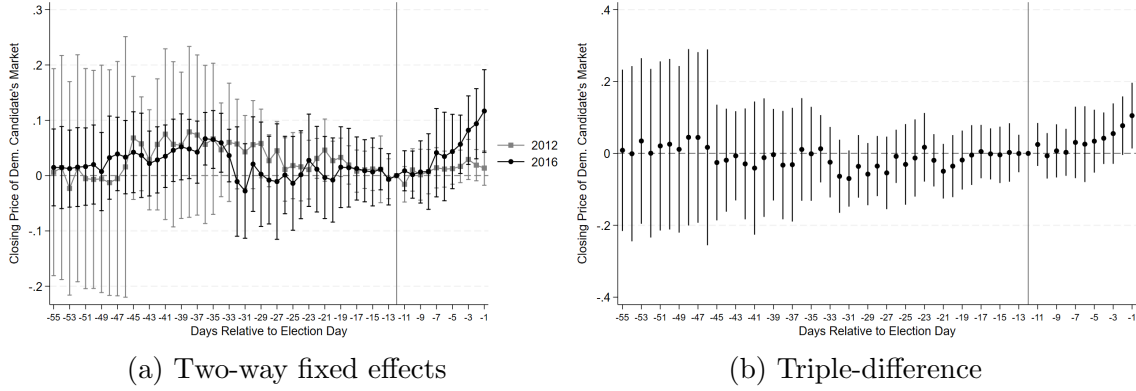


Figure A6: There are no significant pretrends in daily closing price (predicted state-level vote share) of the Democratic presidential candidate’s market in 2012 or 2016.

Notes: This figure displays results from two event study specifications. In Panel A, two-way fixed effects models are estimated separately in 2012 and 2016 and displayed together. In Panel B, a triple-difference specification is estimated. Panel A shows results from estimating $y_{sd} = \delta_s + \delta_d + \sum_{k=-55}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_s + \sum_{k=-11}^{-1} \gamma_k \mathbf{1}\{d = k\} \times treat_s + \epsilon_{sd}$, where s indexes state and d indexes days relative to Election Day. Note that $treat_s$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 and 0 otherwise. Panel B shows results from estimating $y_{sdt} = \delta_{st} + \delta_{sd} + \delta_{dt} + \sum_{k=-55}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_{st} + \sum_{k=-11}^{-1} \gamma_k \mathbf{1}\{d = k\} \times treat_{st} + \epsilon_{sdt}$, where s indexes state, d indexes days relative to Election Day, and t indexes year. Note that $treat_{st}$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 (only turning on in 2016) and 0 otherwise. Each specification is weighted by voting-age population with standard errors clustered at the state level; 95% confidence intervals are displayed. Note that this data is only available for a subset of 21 states in 2016 (9 treatment and 12 control); I thus limit the sample in 2012 to match.

Finally, I use daily survey data from the Cooperative Congressional Election Survey, or CCES, a large-scale survey which interviews voters on their turnout and vote intentions

³⁶Source: <https://www.predictit.org/>.

³⁷The control states represented are Arizona, Georgia, Iowa, Indiana, Maine, North Carolina, Nevada, Ohio, Texas, Utah, Wisconsin, and Wyoming, and the treated states are Florida, Michigan, Missouri, Mississippi, New Hampshire, New York, Pennsylvania, South Carolina, and Virginia.

throughout the election cycle. According to survey documentation, samples are sufficiently large so that survey responses are roughly representative of each state. I aggregate the responses to the state-week level using survey weights and examine intentions to turn out, and conditional on turnout intention, intentions to vote for the Democratic presidential and U.S. House candidates.

Results are presented in Figures A7, A8, and A9. There are no significant pretrends in 2012 and 2016. Interestingly, there is a (non-significant) increase in turnout intention, as well as intention to vote Democrat, in non-early voting states in 2016. It is important to note that all of these responses are conditional on not yet having turned out. Therefore, we would expect higher turnout intentions in states where no voters have had the opportunity to vote yet.

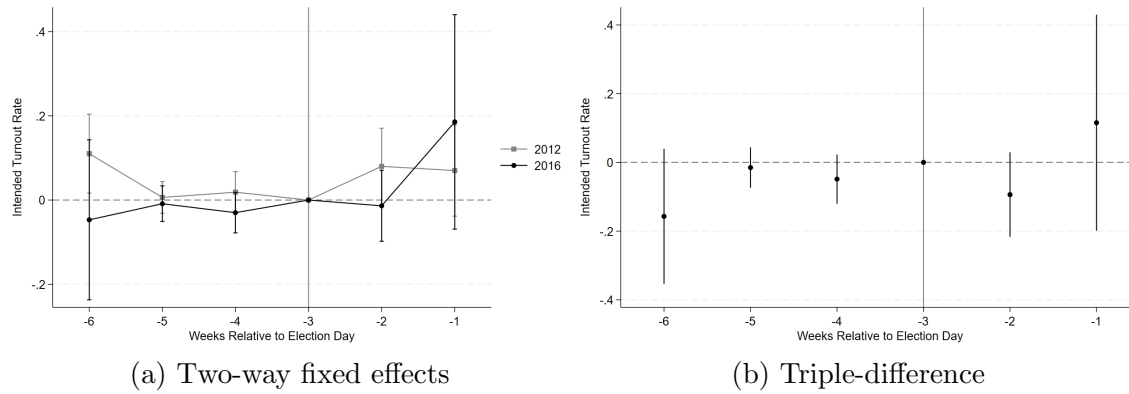


Figure A7: Logged weekly CCES intended turnout rate did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure displays results from two event study specifications. In Panel A, two-way fixed effects models are estimated separately in 2012 and 2016 and displayed together. In Panel B, a triple-difference specification is estimated. Panel A shows results from estimating $y_{sw} = \delta_s + \delta_w + \sum_{k=-6}^{-1} \mu_k \mathbf{1}\{w = k\} \times treat_s + \sum_{k=-2}^{-1} \gamma_k \mathbf{1}\{w = k\} \times treat_s + \epsilon_{sw}$, where s indexes state and w indexes weeks relative to Election Day. Note that $treat_s$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 and 0 otherwise. Panel B shows results from estimating $y_{swt} = \delta_{st} + \delta_{sw} + \delta_{wt} + \sum_{k=-6}^{-1} \mu_k \mathbf{1}\{w = k\} \times treat_{st} + \sum_{k=-2}^{-1} \gamma_k \mathbf{1}\{w = k\} \times treat_{st} + \epsilon_{swt}$, where s indexes state, w indexes weeks relative to Election Day, and t indexes year. Note that $treat_{st}$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 (only turning on in 2016) and 0 otherwise. Weekly average intended turnout rates are calculated for each state using CCES' survey weights so that they are representative of each state. Standard errors are clustered at the state level; 95% confidence intervals are displayed.

Regarding the (non-significant) increase in intention to vote Democrat, it is likely true that each state had a group of voters strongly committed to voting Democrat. In early voting states, many of those voters may have voted early and therefore would not be represented in the survey in later weeks. In non-early voting states, these voters have not yet had the opportunity to vote, and thus would be represented in the survey in later weeks. Therefore,

I interpret these non-significant increases as an indication that marginal voters who had not yet voted in treatment and control states responded to the letter similarly. The difference in state-level responses, then, comes from the fact that voters strongly committed to the Democratic party are no longer in the survey group in control states (as they had already voted early), but are in the survey group in treated states (as they had no chance to vote early).

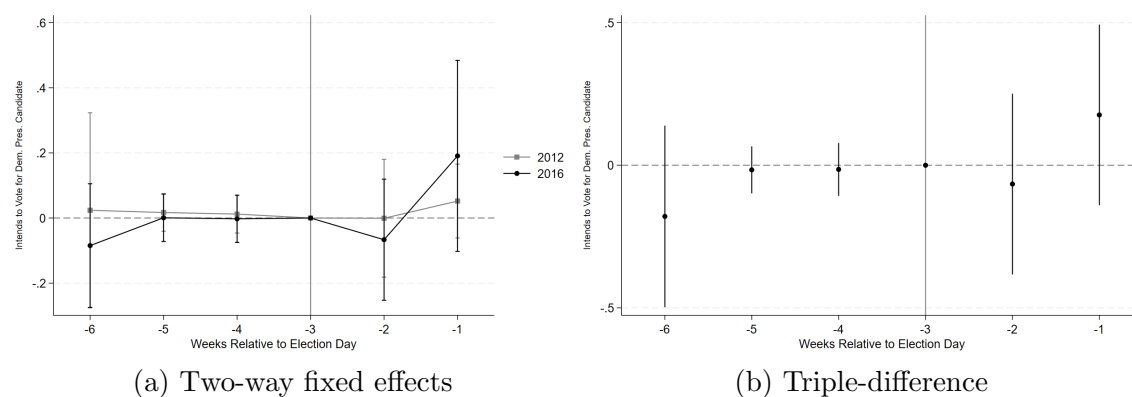


Figure A8: Logged weekly CCES intended Democratic presidential vote share did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure re-estimates equations used in Figure A7 using share of voters who indicated an intention to vote for the Democratic presidential candidate as the outcome variable.

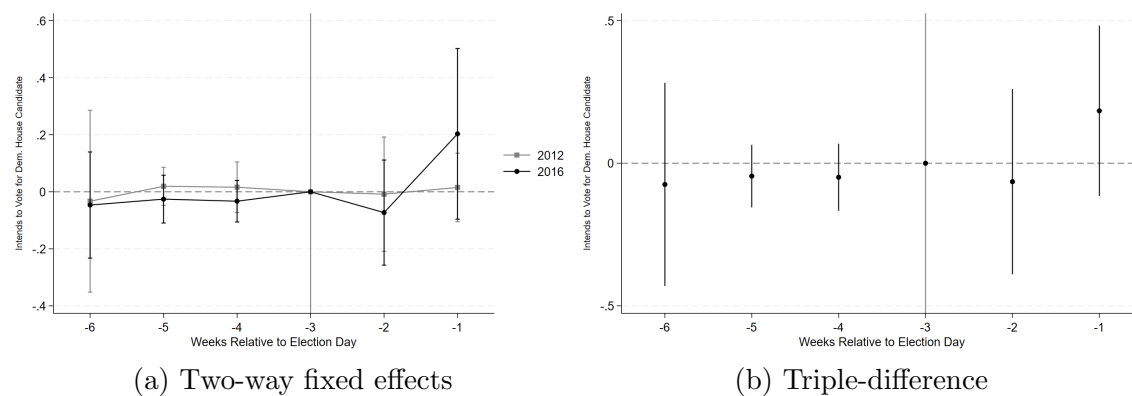


Figure A9: Logged weekly CCES intended Democratic U.S. House vote share did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure re-estimates equations used in Figure A7 using share of voters who indicated an intention to vote for the Democratic U.S. House candidate as the outcome variable.

A.2 Robustness Checks and Additional Specifications

A.2.1 Accounting for White Voters' Education Status

Given the large gap in 2016 among white voters by education [Doherty et al., 2018], it is important to ensure that equation (3) is not capturing a change among white non-college voters and attributing it to the Comey letter (this would be a concern, for example, if non-early voting states had a higher concentration of white voters without a college degree). In Table 1, I show that treatment and control states have similar shares white without a college degree in 2012 (in fact, early voting states have a slightly larger share). I include share white without a college degree in my vector of time-varying controls in Tables 2 and 3. Finally, I re-estimate equation (3) with the inclusion of a term interacting 2016 and share white without a college degree. Results are robust to the inclusion of this term, as shown in Table A1.

Table A1: Results are robust to the inclusion of a term interacting 2016 and share white without a college degree.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	0.008 (0.006)	0.008 (0.006)	-0.024 (0.02)	-0.024 (0.015)	0.036* (0.015)	0.036** (0.015)	-0.035** (0.015)	-0.034** (0.015)	0.032* (0.017)	0.031* (0.016)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.56		0.347		0.618		0.37		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 2 after including a term interacting 2016 and share white without a college degree.

A.2.2 Florida and Oklahoma

In this section, I make a refinement to the design used to estimate Table 2 so that I only compare early voting states to other early voting states. I limit treatment counties to those with early voting open only *after* 10/28 (all counties in Florida and Oklahoma). Then, I re-estimate (3) with the refined treatment group and same control group (all counties in states with early voting open prior to 10/28). This allows me to make direct comparisons between early voting states *only*, rather than relying on comparisons between early voting

and non-early voting states.

Results are in Table A2. Here, there *is* a change on the extensive margin: voters in Florida and Oklahoma, who were more able to respond to the Comey letter, saw an increase in turnout rate of 1.5 percentage points. This is a bit puzzling: if the control group already has a number of voters “locked in” and the rest (who did not vote early) are free to respond, then an increase in turnout does not make much sense. This finding persists when limiting the control group to counties in neighboring states (Arkansas, Georgia, Kansas, New Mexico, and Texas; see Panel B). However, looking at disaggregated turnout of a subset of these neighbors in Table A3, I show that the Comey letter did not lead to a significant change in early turnout rate or Election Day turnout rate. This result appears noisy and is perhaps due to measurement error in the denominator.³⁸

Returning to Table A2, the results show an increase in Republican presidential and state-level vote shares. These results hold (and strengthen) when comparing Florida and Oklahoma only to their neighboring states (Panel B). Broadly, the results remain consistent with those found in my main specification and reported in Table 2.

Table A2: Voters in Florida and Oklahoma were more likely to vote for Republican presidential and down-ballot candidates than voters in states with convenience voting open before 10/28/16.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Panel A: All Control States										
Treat×Post	0.015*** (0.004)	0.013** (0.006)	-0.006 (0.01)	-0.008 (0.017)	0.017* (0.01)	0.018 (0.015)	-0.048*** (0.011)	-0.048*** (0.015)	0.051*** (0.016)	0.051*** (0.015)
N	4180		4180		4180		3814		3814	
Mean	0.555		0.334		0.629		0.354		0.617	
Controls	✓		✓		✓		✓		✓	
Panel B: Neighboring Control States										
Treat×Post	0.012** (0.003)	0.01 (0.009)	-0.03** (0.009)	-0.008 (0.023)	0.031*** (0.007)	0.009 (0.017)	-0.063** (0.02)	-0.027 (0.019)	0.079** (0.032)	0.046*** (0.012)
N	1540		1540		1540		1288		1288	
Mean	0.507		0.298		0.674		0.317		0.66	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 using only counties in Florida and Oklahoma as treatment units. Panel B restricts the set of controls to counties in states neighboring Florida and Oklahoma.

³⁸If voting-age population were growing more quickly in Florida than documented by the SEER measures, this could lead to a false finding of increased turnout rate.

Table A3: The Comey letter led to no significant change in the Election Day or early turnout rates in Florida and Oklahoma counties, as compared to counties in their neighboring states: Arkansas, Georgia, and New Mexico.

	Turnout Rate		Early Turnout Rate		Election Day Turnout Rate	
Treat×Post	0.017*** (0.003)	0.018** (0.005)	0.015 (0.013)	0.018 (0.023)	-0.008 (0.013)	-0.011 (0.026)
<i>N</i>	816		816		816	
Mean	0.512		0.2		0.266	
Controls	✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table disaggregates results from Table A2 Panel B, examining the impact of the letter on turnout by method. Note that the control sample differs from that in Table A2 Panel B: Kansas and Texas do not provide turnout by voting method and are thus dropped from the estimation.

A.2.3 Hurricane Sandy

The first empirical design, summarized in equation (3), takes 2012 as a base year for the difference-in-differences specification. Thus, I assume that any late-election information in 2012 did not affect election outcomes. One potential violation of this assumption comes from Hurricane Sandy, a tropical cyclone which made landfall in the United States around October 29, 2012. Many at the time posited the storm would benefit incumbent Democrat President Obama, who was widely praised for his response. However, as found in Hart [2014], the effect of Hurricane Sandy on the election was “variable and small in magnitude.”

To account for any possible impact of Hurricane Sandy, I re-estimate equation (3) after omitting counties in Connecticut, New Jersey, and New York. These three states had the largest populations in storm surge zones during Hurricane Sandy (Center for International Earth Science Information Network [2012]).³⁹ As such, it is probable that any effect Hurricane Sandy had on political outcomes was largest there. Results are robust to this omission; see Table A4.

³⁹Along with Massachusetts, which is already excluded from my analysis.

Table A4: Results are robust to the omission of Connecticut, New Jersey, and New York.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.005 (0.007)	0.005 (0.007)	-0.02* (0.012)	-0.025 (0.016)	0.032*** (0.011)	0.036** (0.016)	-0.042*** (0.013)	-0.045*** (0.016)	0.039** (0.017)	0.04** (0.017)
<i>N</i>	5600		5600		5600		5218		5218	
Mean	0.561		0.342		0.623		0.364		0.61	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after omitting counties in Connecticut, New Jersey, and New York from the analysis.

A.2.4 Placebo Treatment Dates

It is possible that the central empirical design simply captures changing dynamics throughout the campaign leading up to the election. For example, October 7th was an important day in the 2016 presidential election. Three major events unfolded: *The Washington Post* reported on the 2005 “Access Hollywood” tape, the U.S. government accused Russia of meddling in the election, and WikiLeaks began releasing emails of John Podesta. It could also be the case that Clinton and the Democratic party would have performed worse with “later” versus “earlier” voters, even in the absence of the letter.

To address this concern, I conduct placebo tests where I redefine treatment around a few other key dates during the election: September 26th, the date of the first presidential debate, October 7th, and October 19th, 2016, the date of the final presidential debate. If I find that voters reacted to the placebo “treatment”, then it is likely that the results in Table 2 are reflective of voter preferences shifting over time in response to a number of factors rather than the singular Comey letter “shock.” As can be seen in Table A5, I do *not* find that voters responded to these false treatments, increasing confidence in the attribution of my results to the Comey letter.

Table A5: The results in Table 2 do not hold when treatment is redefined around other major dates in the 2016 election cycle (9/26/16, 10/7/16, 10/19/16), suggesting the results were due to the Comey letter and not simply the larger dynamics of the election.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
First Debate	0.010 (0.019)	0.012 (0.018)	0.022 (0.014)	0.02 (0.021)	-0.023** (0.009)	-0.021 (0.016)	0.014 (0.012)	0.01 (0.014)	-0.007 (0.016)	-0.004 (0.017)
Oct. 7, 2016	0.001 (0.013)	0.00 (0.012)	0.021 (0.014)	0.023 (0.019)	-0.016 (0.014)	-0.02 (0.018)	0.007 (0.019)	0.008 (0.022)	0.00 (0.02)	-0.001 (0.022)
Final Debate	0.001 (0.006)	0.005 (0.007)	-0.006 (0.016)	-0.014 (0.021)	0.011 (0.016)	0.02 (0.021)	-0.002 (0.019)	-0.012 (0.023)	0.00 (0.022)	0.009 (0.023)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.56		0.347		0.618		0.37		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after redefining treatment: counties without early voting open before 9/26/16 (“First Debate”), 10/7/16 (Access Hollywood, WikiLeaks, etc.), 10/19/16 (“Final Debate”) are considered treated in each respective regression. Those with early voting open ahead of those dates are controls.

A.2.5 Placebo Elections

Similarly, one might be concerned that my main empirical approach captures differential trends across treatment and control states over a number of elections. That is, perhaps it is the case that non-early voting states were becoming more right-leaning throughout the 2010s. To address this, I conduct two additional placebo tests. Keeping the definition of treatment the same, I re-estimate equation (3) comparing 1) the 2012 and 2008 elections (with 2012 as the “treated” election) and 2) the 2014 and 2018 elections (with 2018 as the “treated” election).⁴⁰ Results are shown in Tables A6 and A7. I find no effects of treatment in either of these specifications, lending support to the idea that I am estimating the effect of the Comey letter itself and not broader political trends in treatment and control states.

⁴⁰Note that Washington, D.C., Rhode Island, and Washington all made changes to their early voting laws between 2008 and 2012. I omit them from the first analysis. I omit Colorado, Massachusetts, Minnesota, and Montana in both analyses for consistency with the original specification.

Table A6: The results in Table 2 cannot be reproduced when comparing the 2008 and 2012 elections, suggesting the results were due to the Comey letter and not broader trends over several elections in the 2010s.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	-0.002 (0.006)	-0.004 (0.006)	0.005 (0.007)	0.015 (0.013)	-0.003 (0.008)	-0.012 (0.012)	0.016 (0.018)	0.026 (0.02)	-0.009 (0.017)	-0.019 (0.02)
<i>N</i>	5676		5676		5676		5436		5436	
Mean	0.567		0.395		0.589		0.424		0.559	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the specification used in Table 2, using 2012 as the “post” election and 2008 as the “pre” election.

Table A7: The results in Table 2 cannot be reproduced when comparing the 2014 and 2018 elections, suggesting the results were due to the Comey letter and not broader trends over several elections in the 2010s.

	Turnout Rate		Dem. House Vote Share		Rep. House Vote Share		Dem. Senate Vote Share		Rep. Senate Vote Share	
Treat×Post	-0.003 (0.015)	-0.005 (0.015)	-0.006 (0.016)	-0.009 (0.016)	0.008 (0.016)	0.011 (0.014)	0.034 (0.059)	-0.006 (0.038)	-0.021 (0.056)	0.02 (0.037)
<i>N</i>	5758		5385		5385		3536		3536	
Mean	0.418		0.346		0.633		0.36		0.605	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the specification used in Table 2, using 2014 as the “post” election and 2018 as the “pre” election. As presidential elections did not take place in these years, I report the vote shares for U.S. House and Senate races separately, rather than an average of House, Senate, and gubernatorial vote shares.

A.2.6 Battleground States

In the following section, I present results separately for battleground and non-battleground states. Here, I categorize all states deemed to have competitive presidential races by Cook Political Report as battleground states. Figure A10 displays variation in convenience voting laws for U.S. states in 2016, broken out by battleground status. Recall that there is no effect

of treatment on states' battleground status (Figure A1), meaning that the results in Table 2 are not driven by differentially competitive presidential races in treatment or control states.



Figure A10: This map displays variation in convenience voting laws for U.S. states in 2016, broken into battleground and non-battleground states.

It is plausible that the letter's effect differed in presidential battleground and non-battleground states, as the level of campaign activity in the former is likely much higher. However, this is not to say that coattail effects are of no significance in non-presidential battleground states. Note that in 2016, 12 of 23 battleground U.S. House races and 3 of 9 battleground Senate races were held in non-presidential battleground states [Ballotpedia, e,f]. Although voters in these races likely did not believe their vote would be pivotal in the presidential race, they still may have made assumptions about the quality of down-ballot candidates based on their impressions of their top-ballot co-partisans. This is especially likely in a presidential race, where low-propensity (and less-informed) voters are more likely to vote.

In Table A8, I re-estimate results from Table 2 for 2016 battleground states (Panel A) and non-battleground states (Panel B) separately. Note that the effects on Republican presidential vote share, Democratic down-ballot vote share, and Republican down-ballot vote share are stronger in non-battleground states, but effects on these outcomes do not significantly differ by battleground status.

However, the effects on turnout and Democratic presidential vote share do significantly differ in battleground and non-battleground states. In battleground states, exposure to the letter increased turnout by 3 percentage points and led to a very small (and non-significant) increase in Democratic presidential vote share (overall, the letter led to a non-significant increase in Republican presidential vote margin of 1.3 percentage points in battleground

states). One interpretation of these results is that in presidential battleground states, the letter was more likely to mobilize right-leaning low-propensity voters than it was to discourage left-leaning low-propensity voters and, perhaps, induced vote-switching away from third-party candidates. This interpretation is consistent with the idea that perceived closeness of an election leads to higher turnout, although evidence is mixed [Bursztyrn et al., 2023, Gerber et al., 2020]. While the mechanisms may have differed between battleground and non-battleground states, the down-ballot effects of the Comey letter did not.

Table A8: The results in Table 2 appear to be driven by non-battleground states, although the difference between the coefficients for Republican presidential vote share, Democratic down-ballot vote share, and Republican down-ballot vote share are not statistically significant.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Panel A: Battleground States										
Treat $\times$ Post	0.03*** (0.007)	0.029** (0.011)	0.003 (0.015)	0.003 (0.013)	0.016 (0.02)	0.015 (0.019)	-0.02 (0.018)	-0.019 (0.017)	0.016 (0.021)	0.016 (0.02)
N	1664		1664		1664		1538		1538	
Mean	0.59		0.362		0.604		0.376		0.601	
Controls	✓		✓		✓		✓		✓	
Panel B: Non-Battleground States										
Treat $\times$ Post	-0.001 (0.008)	-0.002 (0.008)	-0.031** (0.011)	-0.037* (0.019)	0.039*** (0.011)	0.046** (0.019)	-0.037** (0.014)	-0.041** (0.019)	0.033* (0.018)	0.038* (0.02)
N	4118		4118		4118		3862		3862	
Mean	0.548		0.341		0.624		0.368		0.606	
Controls	✓		✓		✓		✓		✓	
Difference in Coefficients										
p-values	0.003	0.024	0.077	0.075	0.291	0.25	0.469	0.377	0.524	0.428

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 2 using 2016 battleground states (Panel A) and non-battleground states (Panel B) only. The final row in the table gives the p -values from testing if the coefficients differ significantly using the Z -statistic $Z = \frac{(\beta_a - \beta_b)}{\sqrt{SE(\beta_a)^2 + SE(\beta_b)^2}}$ [Clogg et al., 1995].

A.2.7 Non-Continuous Specification

In this section, I estimate a non-continuous version of (4). In an attempt to ensure that a sizable proportion of the “controls” (early voters) in 2016 cast their ballots ahead of the letter’s release, I limit the sample to states with more than 5 days of early voting ahead of

October 28th, as motivated by the high early voting turnout rates in those states, shown in Figure 4. Further, I weight regressions by total votes cast, as I expect counties with more votes cast during the early voting period (versus less) to be more representative of voters’ preferences pre-letter. Here, I cluster at the county level, as treatment is no longer determined by states’ early voting policy. I find a significant decrease in Democratic presidential vote share.

Table A9: Results in Table 4 are robust to a non-continuous specification.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share	
Treat×Post	-0.012** (0.005)	-0.009* (0.005)	0.0002 (0.005)	-0.002 (0.005)
<i>N</i>	1432		1432	
Mean	0.385		0.588	
Controls	✓		✓	

Standard errors are clustered at the county level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-period level difference-in-differences specification, weighted by total votes cast: $y_{psct} = \alpha + \delta T_p + \gamma P_t + \eta F_c + \beta(T_p \times P_t) + \epsilon_{psct}$, where T_p takes a value of 1 in the Election Day voting period and 0 in the early voting period, and P_t takes a value of 1 in 2016 and 0 in 2012. County fixed effects are given by F_c . I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor’s degree or higher, and share white without a college degree. Presidential vote shares are proportions of total presidential votes cast. Only counties with at least five early voting days occurring before the letter’s release are included.

A.2.8 North Carolina and Nevada

As discussed in Section 4.4, counties in North Carolina had wider discretion over hours and locations for early in-person voting in 2012 and 2016. Although there is no evidence of systemic effects on turnout stemming from changes in total hours or number of sites [Walker et al., 2019] and no reason to expect these changes would have differential effects for Democrats and Republicans around the day of the letter’s release, one may be concerned that results in Figure 5 are attributable to potential changes in early voting availability rather than the Comey letter. Additionally, one might question if the significant decline in Democrat turnout found after the letter’s release is solely due to a regression to the mean after higher turnout in the first few days of the early voting period (perhaps due to mobilization in response to Republicans’ attempts to cut early voting).

Here, I present the results from equation (5) in Figure 5 in Table A10 Panel A1 and re-estimate them after omitting the first three days of early voting in Table A10 Panel A2. While the treatment effect does decrease, results remain significant and qualitatively similar: after the letter’s release, we observe a decline in turnout by registered Democrats relative to 2012. I provide additional evidence to bolster the findings in North Carolina by leveraging disaggregated early turnout data provided by Nevada. In 2016, Nevada had a fourteen-day early voting period that lasted between October 22 and November 4. The state reports partisan turnout during Week 1 (10/22/16-10/28/16) and Week 2 (10/29/16-11/4/16) separately, allowing me to compare “Week 1” and “Week 2” early voters in 2012 and 2016.⁴¹ Note that 10/28 is included in the “control” group here as data is reported at a weekly, not daily, level. As such, the results I find likely represent an underestimate of the letter’s true effect.

Using week-county-year observations, I estimate a difference-in-differences specification in Nevada, comparing Week 1 and Week 2 voters in 2016 and in 2012. The identifying assumption here is that, absent the Comey letter, Week 1 and Week 2 voters would have trended similarly in terms of the outcomes of interest. Again, this design also requires that any late-election information in 2012 did not affect political behavior in Nevada. I estimate

$$y_{wct} = \alpha + \delta T_w + \gamma P_t + \beta(T_w \times P_t) + \delta_c + \epsilon_{wct}, \quad (11)$$

where w indexes week, c indexes county, and t indexes year. Here, T_w takes a value of 1 in Week 2 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. I include county fixed effects δ_c to account for any time-invariant cross-county differences in early voting administration. Outcomes of interest are turnout by party registrants (i.e., voters registered as Democrats, etc.) as a share of total early votes. I weight by voting-age population and cluster at the county level as in (5).

Table A10 Panel B shows results from the complementary Nevada design. Due to the small number of clusters (17 counties), I implement a wild cluster bootstrap, estimated in Stata with the “boottest” command [Roodman et al., 2019]. This method does not assume normality and as such, standard errors are not estimated. Instead, I report p -values in parentheses and 95 percent confidence intervals in square brackets. Here, I find a decrease of 0.9 percentage points in the share of early ballots cast by registered Democrats. This is likely an underestimate of the letter’s effect, as the date of its release is included in the

⁴¹In 2012, Week 1 refers to turnout between 10/20/12 and 10/26/12, and Week 2 refers to turnout between 10/27/12 and 11/2/12. In both 2012 and 2016, early voting begins 18 days ahead of Election Day and concludes 4 days ahead of Election Day.

“control” group due to data limitations.

Table A10: After the Comey letter was sent, the share of early ballots cast by registered Democrats in Nevada and North Carolina decreased, and the Republican early voting share in North Carolina increased.

	Registered Dem. Early Turnout Share	Registered Rep. Early Turnout Share	Registered Ind. Early Turnout Share
Panel A1: North Carolina			
Letter	-0.023*** (0.006)	0.023*** (0.007)	-0.00002 (0.002)
<i>N</i>	2884	2884	2884
Mean	0.469	0.311	0.22
Panel A2: North Carolina – first three days omitted			
Letter	-0.014*** (0.004)	0.013** (0.006)	0.002 (0.002)
<i>N</i>	2334	2334	2334
Mean	0.457	0.319	0.224
Panel B: Nevada			
Treat × Post	-0.009 (0.099) [-0.02997, 0.0036]	0.001 (0.92) [-0.0083, 0.0264]	0.009 (0.362) [-0.0173, 0.0223]
<i>N</i>	68	68	68
Mean	0.277	0.532	0.191

In Panel A, standard errors are clustered at the county level and listed in parentheses. In Panel B, I cluster at the county level and list wild cluster *p*-values in parentheses, with wild cluster bootstrap 95 percent confidence intervals are given in square brackets.

In Panel A: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Panel A shows estimates from a two-way fixed effects specification, weighted by voting-age population: $y_{c dt} = \delta_d + \delta_t + \beta Letter_{dt} + X_{ct} + \epsilon_{c dt}$, where c indexes county, d indexes days relative to Election Day, and t indexes year. Note that $Letter_{dt}$ is a dummy variable equaling 1 for observations at most eleven days ahead of Election Day and 0 otherwise (and only turning on in 2016). I include a vector of county-level early voting characteristics, X_{ct} . Panel A1 includes the first three days of the early voting period; Panel A2 excludes them. Panel B shows results from a county-period level difference-in-differences specification, weighted by voting-age population: $y_{wct} = \alpha + \delta T_w + \gamma P_t + \beta(T_w \times P_t) + \delta_c + \epsilon_{wct}$, where T_w takes a value of 1 in Week 2 and 0 otherwise, P_t takes a value of 1 in 2016 and 0 in 2012. County fixed effects are given by δ_c . In both panels, registered Democrat early turnout share is the number of registered Democrats who cast a ballot early divided by the number of early ballots cast; registered Republican and Independent early turnout shares are defined similarly for registered Republicans and voters not registered as Republicans or Democrats, respectively.

A.2.9 Instrumented Difference-in-Differences Results on U.S. House Vote Shares

The results in Table 4 demonstrate the impact of a one percentage point increase in Republican presidential vote share on partisan down-ballot vote shares. I calculate these down-ballot vote shares by calculating the average of Republican or Democratic U.S. House, Senate, and gubernatorial vote shares. However, much of the presidential coattails literature uses only U.S. House vote shares as outcomes – likely because, unlike senators and governors, U.S. House representatives face re-election every two years.

Table A11 presents results from the DDIV model where the outcomes of interest are partisan U.S. House vote shares. I find that a one percentage point increase in Republican presidential vote share leads to a 0.51 percentage point increase in Republican U.S. House vote share.

Table A11: An increase in Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) U.S. House vote shares.

	(1)	(2)
Rep. U.S. House Vote Share	0.514** (0.254)	0.548** (0.224)
Dem. U.S. House Vote Share	-0.49* (0.291)	-0.527* (0.303)
First-Stage (Rep. Pres. Vote Share)	0.035*** (0.013)	0.037** (0.016)
N	4910	4910
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 4 using only U.S. House vote shares as the outcome of interest. I drop observations where only one major party competed.

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